

**Adjoint sensitivities of sea level:
Mechanisms for changes of variability in
the Indian Ocean Sector of the ACC under
strong warming scenario**

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November, 2022
Hamburg, Germany

Accepted as Master's thesis
from the Department of Earth System Sciences at the University of Hamburg

Date of the defense:

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Hamburg, January 5, 2023.

Abstract

We estimate the relative contribution of surface wind stress and buoyancy components to the long periods (≥ 3 years) and large scales ($\geq 300\text{km}$) sea-level variability under the global warming scenario in the Southern Ocean Indian Sector. The underlying mechanisms of sea-level variability are investigated using a quasi-global configuration of the Massachusetts Institute of Technology general circulation model and its adjoint model.

We define the objective function as a three-year mean sea level height (SSH) anomaly averaged over a box region. Its sensitivities to atmospheric forcings and earlier states are estimated using the adjoint model. In chronological order, the sensitivities to the surface wind stress in the Southern Ocean propagates westward, which are advected by mean Antarctic Circumpolar Current (ACC) flow. In contrast, the sensitivities to the buoyancy components propagate eastward along the ACC mean flow.

Projecting the forcing variability into the adjoint sensitivity fields reveals that the large-scale sea level variation is primarily influenced by the zonal wind stress over the Southern Ocean. This accounts for 48% of the total variance.

The local wind from south Australia has the largest contribution. It promotes Ekman pumping which is shifted westward by the westward baroclinic Rossby wave incited by surface forcing. While the non-local wind from the equator contributes a small fraction, the effects take 6-9 years to arrive at the region of interest.

The surface heat and freshwater flux influence approximately a third of sea level variation, which the steric effects can explain.

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Chapter 1

Introduction

This chapter introduces the thesis topic briefly. First, we explain the motivation for studying large-scale ($\geq 300\text{km}$), low-frequency (≥ 3 years) regional sea level height (SSH) variability in the Southern Ocean Indian Sector. Second, we review the state-of-the-art in regional sea level studies. Then we illustrate key ocean dynamics in the region of interest. Finally, we state the research objectives and summarize the thesis outline.

1.1 Motivation

Understanding sea level change and its variability is a critical topic in climate science with far-reaching implications for coastal communities and ecosystems. However, studying sea level change is challenging owing to the complexity of the climate system. As a result, there has been much uncertainty in understanding sea level variability and projecting sea level rise. Continued research into sea level science in response to climate change is required to improve risk assessment, climate mitigation, and adaptation (Church et al., 2007, 2013).

The scientific community has traditionally concentrated on the global mean sea level (GMSL). However, sea level change is not uniform globally but exhibits substantial regional patterns in space and time that may deviate significantly from the global mean (Stammer et al., 2013). Furthermore, sea level changes regionally due to the heterogeneous physical processes at work. For instance, according to the sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC AR6), sea levels rose most rapidly in the Western Pacific from 1993 to 2018 and most slowly in the Eastern Pacific (Fox-Kemper et al., 2021).

The difference between GMSL and regional sea level is also reflected in future sea level change projections. Slangen et al. (2014) suggests regional sea level variations in the twenty-first century could be up to 30% above and 50% below the global mean.

The IPCC AR6 claims that regional sea level rise will be within $\pm 20\%$ of the median projected GMSL change (Fox-Kemper et al., 2021).

It is, therefore, essential to emphasize regional sea level change and its variability. Regional sea level changes reflect large-scale atmospheric and oceanic changes. Although their underlying driven mechanism may vary from region to region (Yin et al., 2010). Therefore, understanding the causal attribution and forcing mechanism is critical for predicting regional sea level change. It is worth mentioning that the scenario-dependent projection of sea level change raises significant societal concerns but there remains a large range of uncertainty by the end of the twenty-first century (Slangen et al., 2014; Stammer et al., 2019).

One of the critical regions with limited understanding and high uncertainty is the Southern Ocean. This plays a crucial role in the climate system. The Southern Ocean south of 30°S accounts for 40% of ocean carbon uptake and absorbs 75% of total anthropogenically generated heat despite accounting for 30% of ocean area (Frölicher et al., 2015).

However, the Southern Ocean has been undergoing rapid changes in recent decades, with potentially wide-ranging consequences on global ocean circulation and the climate system. Remarkably, the westerly wind has been strengthening and shifting poleward, associated with a positive trend of the southern annular mode (SAM) (Marshall, 2003; Frankcombe et al., 2013). The Southern Ocean has been warming since the 1950s, absorbing more heat than any other ocean on Earth (Gille, 2002). It has been acted as the most prominent heat reservoir globally (Cheng et al., 2022). The changes in surface momentum and buoyancy flux alter the ocean's temperature and salinity physically (Swart et al., 2018).

Although satellite altimeters have greatly improved the observation, in-situ measurements are sparse in the Southern Ocean. As a result, previous efforts have centered on ocean models (Meijers et al., 2012; Zika et al., 2013). A few satellite-based research have focused on the high-frequency part of the sea level variability due to relatively short satellite records at the time (Fu, 2003).

Regardless, satellite altimeter observations and model studies have revealed several distinct barotropic modes in the Southern Ocean, such as the Australia-Antarctic Basin between Australia and Antarctica (Fig 1.1). In these regions, high-frequency SSH variability are associated with wind stress, ACC variability, and bottom topography (Fukumori et al., 1998; Webb and de Cuevas, 2002; Fu, 2003; Vivier et al., 2005; Weijer, 2010; Forget and Ponte, 2015). However, the large-scale ($\geq 300\text{km}$), low-frequency (≥ 3 years) sea level variability in the Southern Ocean has received little attention.

Recently, a statistical analysis based on the outputs of Max Planck Institute Grand Ensemble (MPI-GE, Maher et al. (2019)) discovered a significant change in SSH variability in the Southern Ocean Indian Sector (S.D. Nandini-Weiss et al., submitted to J. Geophys. Res. Ocn). Specifically, the interannual SSH off Southwest of Australia

[48-56°S, 78-123°E] is more variable under the Representative Concentration Pathway 8.5 scenario (RCP 8.5, 2081-2100) compared to the historical period (1986-2006). In addition, they emphasize the need for the following research into SSH variability in the same region on an annual to decadal timescale. This indicates that the mechanism remains unknown.

Furthermore, Giarolla and Matano (2013) examined the main modes of SSH variability in the Southern Ocean using empirical orthogonal function (EOF) analysis. The results imply that the spatial extent of sea level modes in the high-frequency part of the variability spectrum may also be evident in the low-frequency component.

Those mentioned above inspired our research on the large-scale, low-frequency sea level variability in the Southern Ocean Indian Sector (off Southwest of Australia/Australia-Antarctic Basin), where a dominant mode of either high or low-frequency variability prevails. The current study employs an inverse modeling technique with a cutting-edge ocean model to investigate the drivers of sea level variation at low-frequency (≥ 3 years). Understanding the sea level variability and its driven mechanism in this section is essential to understanding the role of air-sea flux and ocean circulation variability in the Southern Ocean.

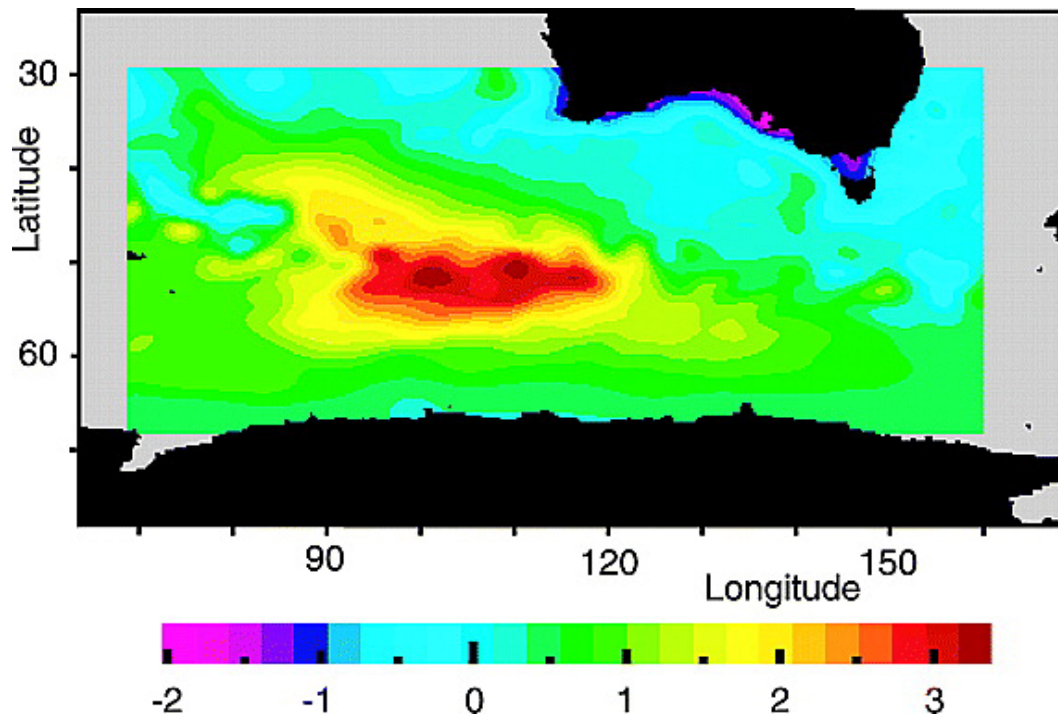


Figure 1.1: The first EOF mode of SSH in the Southern Ocean Indian sector, based on a model driven by high-frequency wind (Webb and de Cuevas, 2002). The first mode accounts for 27% of total SSH variance, which is similar with patterns seen in both satellite and other models. The mode is bounded by the Kergulen Plateau to the west, the Antarctic Continental Shelf to the South, and the Southeast Indian Ridge to the north and east, with a roughly elliptical core extending from 85°E to 111°E and from 44°S to 56°S.

1.2 Regional sea level

Sea level rise is widely acknowledged as one of the most consequential impacts of anthropogenic climate change and is a subject of great importance to society. However, communicating about sea-level change frequently causes ambiguity among the scientific community and the public due to different concepts and terminology (Gregory et al., 2019).

One must first understand the spatial scales of sea level change. GMSL rise is typically on the global scale, primarily caused by thermal expansion and mass loss from glaciers and ice sheets. On the other hand, the extreme sea level, which frequently causes public concern, occurs on the local scale. It involves astronomical tides, storm surges, and coastal flooding. However, it is necessary to stress that the primary concern of the present study is on the regional scale, with distances ranging from tens to hundreds of kilometers (Fox-Kemper et al., 2021).

The temporal scale of sea level change must also be considered. The SSH can fluctuate over a wide range of timescales, from relatively short-lived events like waves and storm surges to long-term changes associated with climate modes. For instance, isostatic adjustment is significant on climate timescales (Fox-Kemper et al., 2021). Therefore, specific physical processes on specific time scales may be primarily responsible for the study of interest. Figure 1.2 depicts a schematic representation of the temporospatial scales of ocean variability (Chelton, 2001).

The regional sea level changes can be summarized into two major components (Stammer et al., 2013): (1) dynamic regional sea level changes that arise from variability in the ocean circulation, redistribution of mass, heat, and freshwater, and are associated with climate modes. e.g., changes in ocean temperature and salinity affect sea level through changes in volume and density; wind stress drives mass circulation via Ekman pumping, planetary waves, and coastal upwelling. (2) static regional sea level change in response to ocean loadings. e.g., the ocean adjusts to changes in surface atmospheric pressure, and changes in earth gravity, rotation, and viscoelastic solid earth deformation due to surface mass redistribution (see Fig1.3).

It is unnecessary to consider all processes when discussing regional sea level variability over relatively short timescales. While ice mass loss has a global impact, its influence on regional sea level change may act on decadal to century timescales in most parts of the world. Ice-induced sea level change may be less significant than regional dynamical processes at interannual timescales. Regional SSH variability in most parts of the oceans primarily arises from the ocean's response to atmospheric forcing, especially wind stress fields, surface heat, and freshwater fluxes (Zhang et al., 2022).

By definition, ocean dynamic sea level is the local height of the sea surface above the geoid, with a global mean of zero. The local SSH anomaly may be positive or negative relative to the global mean, jointly determined by ocean dynamics and density change (Gregory et al., 2019; Fox-Kemper et al., 2021).

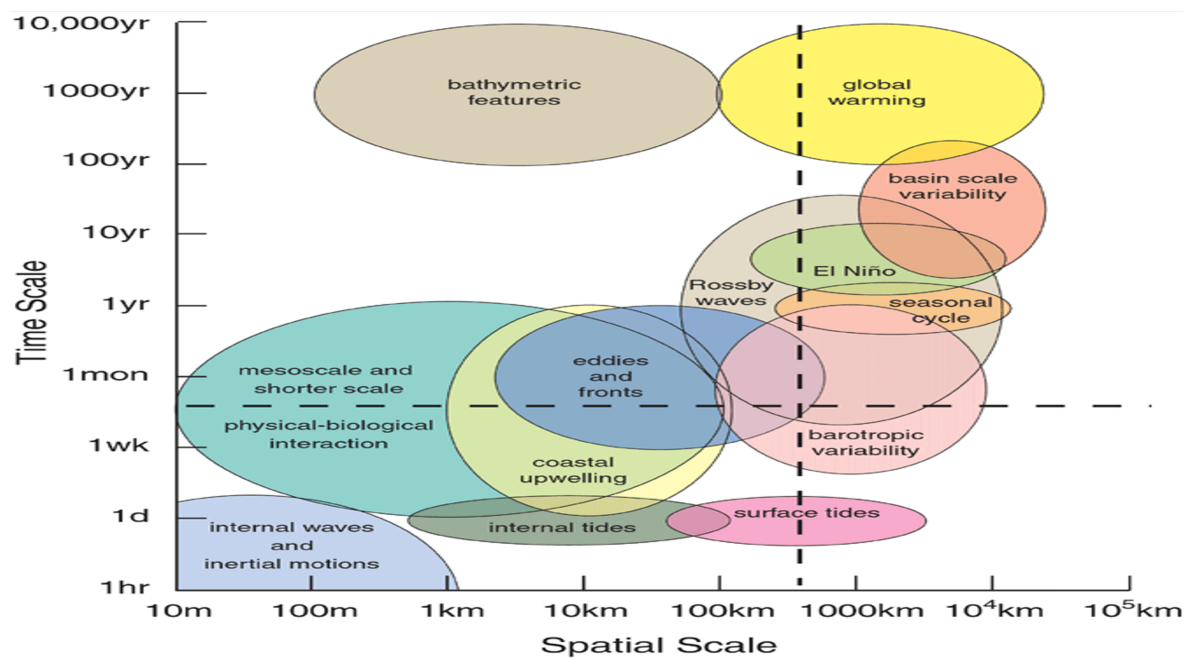


Figure 1.2: Temporal and spatial scales of oceanic variability (Chelton, 2001).

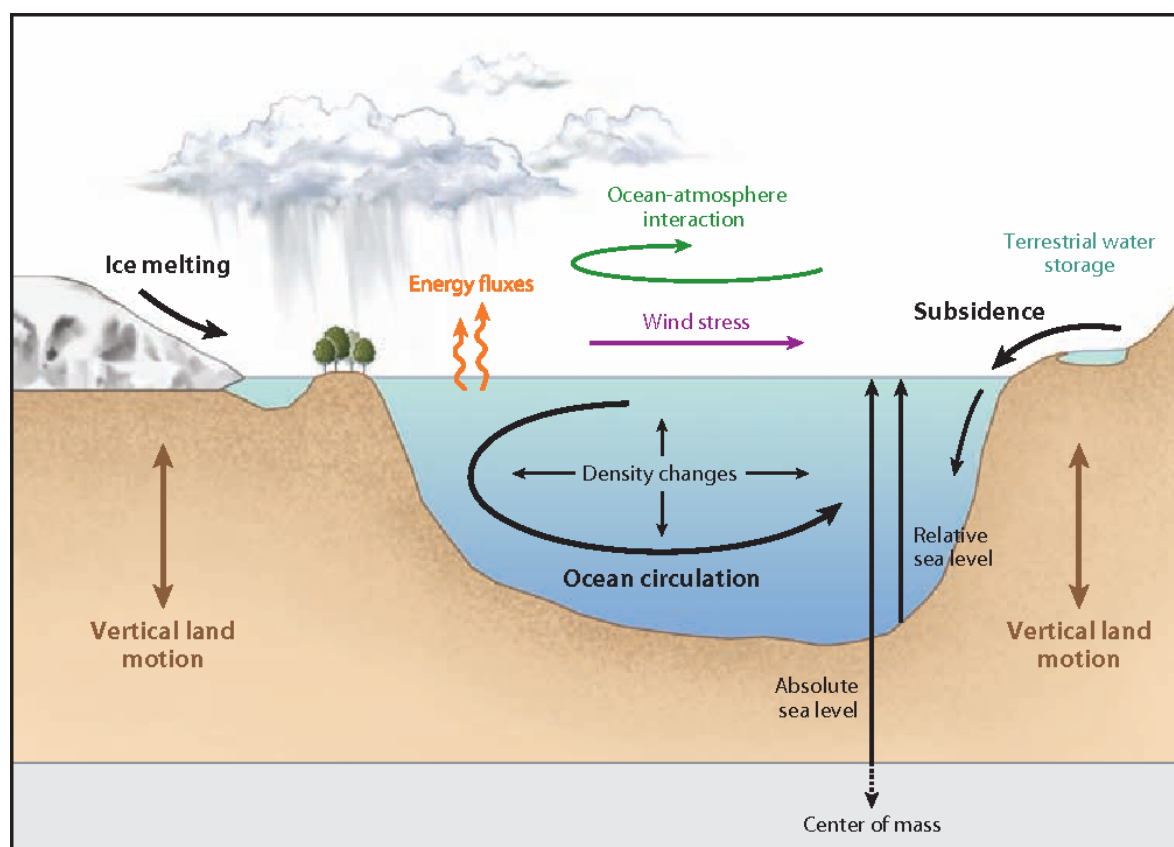


Figure 1.3: Processes that influence regional sea level variation (Stammer et al., 2013).

1.3 Region of interest

The Southern Ocean has been recognized as a critical region for global circulation, ocean carbon uptake, and sea level changes due to many complex interacting physical elements, such as the Meridional Overturning Circulation (MOC), ACC, mesoscale eddies, the Southern Annular Mode (SAM), the shifting Westerly winds, ocean warming and melting ice sheets, peculiar bottom topography, and lateral boundary conditions (Frankcombe et al., 2013; Meijers et al., 2012).

Figure 1.4 depicts the principal dynamics of the Southern Ocean. At the surface ocean, the westerly-driven Ekman transport flows northward. The latitudinal variation of the westerly winds promotes divergence in the Ekman transport, which causes water from the deep ocean to rise to the surface. Upwelled water then divides into two streams: water that reaches the surface near Antarctica is cooled and sinks into the dense bottom water, while water in the north is heated and flows northward along the Ekman transport. Most significantly, as the most significant ocean current, the prevailing ACC flows from west to east through the channel, linking the ocean basins (Morrison et al., 2015).

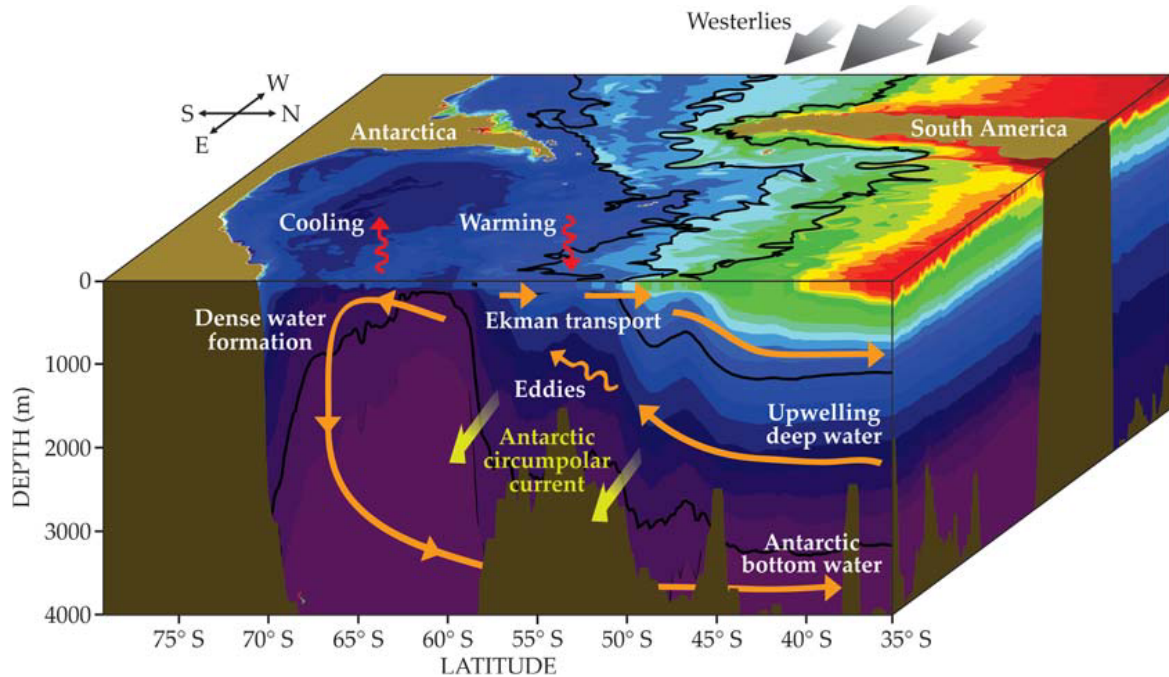


Figure 1.4: Key dynamics in the Southern Ocean (Morrison et al., 2015)

There are two different mechanisms causing local dynamical sea level change. The first is dynamical sea level variation caused by wind-induced Ekman processes or large-scale waves. The second is, steric sea level changes. It arises from ocean surface heating/-cooling and freshening, which lead to density changes and water mixing/transport due to pressure differences. When Ekman transport, ACC, eddies, thermodynamics, to-

pography, and other factors interact, regional sea level change and variability become more challenging to separate (Rintoul, 2018).

The mechanism of Southern Ocean sea level change has been under debate. With the help of CMIP models (Coupled Model Intercomparison Project), Bouttes et al. (2012) concluded that the projected sea level change in the Southern Ocean is mainly due to wind stress change. In contrast, Frankcombe et al. (2013) further investigated the ocean dynamics using perturbed Southern Ocean wind forcing experiments on an eddy-permitting ocean model. They suggest that the thermosteric effect is primarily responsible for the regional sea level change.

The lack of quality-controlled observations has hindered research in the Southern Ocean, even though satellites have been deployed in recent decades. Seasonally formed sea ice complicates satellite observation. Over a short temporal range, both satellites and models have provided insights into regional sea level change and variability in the Southern Ocean. For example, analysis of TOPEX/POSEIDON altimeter data and global ocean model simulations have identified regions with significantly increased SSH variability. These regions include the Australia-Antarctic Basin and the Bellingshausen Basin west of the Drake Passage (Chao and Fu, 1995; Vivier et al., 2005).

One of the most commonly studied regions is the Southern Ocean Indian Sector (also referred to as the Australia-Antarctic Basin in some publications), where satellite altimeter observations and model studies have documented a distinct barotropic mode (Fukumori et al., 1998; Webb and de Cuevas, 2002; Fu, 2003; Vivier et al., 2005; Weijer, 2010; Forget and Ponte, 2015). This region exhibits a broad region of long wavelength elevated intraseasonal SSH variability, featuring a trapped geostrophic mode (Webb and de Cuevas, 2002) (see Fig1.1). Furthermore, (Ponte and Piecuch, 2014) discovered enhanced large-scale bottom pressure variability in nearly the same region, correlated with sea level variability.

Those studies have focused on high-frequency sea level variability, while the low-frequency component has received little attention. However, inspired by a recent statistical analysis using the newly-developed ensemble model that discovered this obvious pattern, we are eager to investigate the low-frequency sea level variability in such a dynamic region. Consequently, we should narrow our focus even further to a smaller region characterizing the low-frequency SSH variability in the Indian sector of the Southern Ocean.

The modes of variability characterize the Southern Ocean as a trend of increasing SSH. As a result, it is worthwhile to look into its underlying mechanism, variability, and predictability. In light of current efforts to understand low-frequency variability, more attention is required due to changing atmospheric forcing and potential sea level change. Understanding large-scale, low-frequency regional sea level variability is essential to understanding Southern Ocean changes, particularly in the face of climate change.

1.4 Thesis objectives and outline

This research focuses on large-scale, low-frequency sea level variability in response to changes in atmospheric forcings, namely surface zonal wind stress, meridional wind stress, net heat flux, and freshwater flux. Despite the advancement of modeling studies, there has been no quantitative exploration of the sensitivity of large-scale sea level variability under a future warming scenario.

Following recent research on sea level variability, this thesis aims to provide a quantitative understanding of the driving mechanism for low-frequency sea level variability in the Southern Ocean Indian Sector. The Massachusetts Institute of Technology general circulation model (MITgcm) and its adjoint model are ideal for this study, as it has been extensively used for oceanic sensitivity analysis. The primary thesis objectives and the questions we aim to answer are:

- 1) Identify relevant physical processes that affect long-term and low-frequency regional sea level variation in the Southern Ocean Indian Sector.
- 2) Quantify the relative importance of various atmospheric forcings under the global warming scenario that contribute to sea level variation.
- 3) Identify how the large-scale SSH anomaly in the Southern Ocean Indian Sector can be optimally excited, which is a typical signal responsible for low-frequency SSH variability. For instance, how does a change in each of the forcings affect the excitement of an SSH anomaly?

The first chapter introduced the topic and objectives. We have outlined the rationale for studying regional sea level variation in the area of interest and reviewed the relevant literature.

The background information of the inverse approach is described in detail in Chapter 2, including an explicit explanation, the simplified mathematical derivation, and its application in oceanography. In addition, we apply an EOF analysis to narrow down the region of interest for this study.

In chapter 3, we evaluate the model configuration and experimental designs. Primarily, we explain the choice of the objective function for the adjoint model and sensitivity-based forward perturbation experiments.

Chapter 4 presents the results of the forward and adjoint solutions. Initially, we investigate the forward solution by considering several crucial factors. Then, the sensitivity pathway, interpretation, and response to forcing are described. Finally, we demonstrate how the perturbation experiment can be utilized to validate the underlying mechanism.

Chapter 5 discusses the driving mechanism of low-frequency sea level variability, as well as the caveats of the present study.

We summarize the research and present a brief outlook in chapter 6.

Chapter 2

Background

In this chapter, we provide background information on the adjoint modeling concept, procedure, mathematical derivation, applications, and limitations. We emphasize the adjoint sensitivity studies of the ocean state using the Ocean General Circulation Model (OGCM). Adjoint modeling is acknowledged as a powerful tool for sensitivity analysis which requires the calculation of the first derivatives of model outputs to inputs (Errico, 1997).

2.1 The adjoint method

The ocean is a fluid with long-lasting memory. Any change in regional oceanographic quantities may result from the propagation or integration of relevant physical processes that occurred at the remote site long ago. To better understand the variability, forcing mechanism, and predictability of the target quantification (usually called objective function or cost function in adjoint modeling), we need to know its sensitivity to each contributing variable at various times and locations (Heimbach et al., 2011). Due to the large quantity and multidimensional nature in oceanography, we need to consider computational efficiency. The so-called adjoint method has been extensively used as an efficient approach to estimating linear sensitivity.

In sensitivity analysis, the difference in objective function between the disturbed and the undisturbed run would be the sensitivity to the variable if a particular perturbation is imposed on a specific control variable earlier. The adjoint technique assesses the sensitivity of the objective function to all factors at once, commonly known as the gradient or the first derivative. Once the sensitivity of an objective function to relevant variables is quantified, the driving elements can be identified, and the underlying mechanism causing the changes can be discovered more easily (Errico, 1997).

A typical nonlinear ocean model takes a set of inputs and predicts a set of outputs. Initial conditions, boundary conditions, model parameters, and forcing fields are examples of inputs. Predicted oceanic states, model-observation mismatches, or any

scalar quantification of state fields are examples of outputs. Initial, boundary, and forcing conditions, such as surface wind stress, drive an ocean model. During model integration, the model generates ocean state fields such as SSH, etc.

It is possible to study any oceanic quantity from the model output fields, such as Atlantic meridional overturning transport, ocean heat content, model-data misfits, etc.), depending on the research interest. However, we are usually interested not only in the output result but also in how the output depends on control variables at previous times, in other words, the sensitivity to control variables.

The conventional way to derive sensitivity is to perturb a specific input quantity at a specific location and time. The difference of the objective function between the perturbed and the control run can thus be calculated. In this case, one single perturbation results in all the differences in outputs, which are usually redundant except for the target. Furthermore, such forward perturbation experiments require multiple ensembles of simulations that vary in magnitude and temporal-spatial location for each input variable, which is indeed feasible but computationally expensive (See schematic plot, Fig.2.1 a).

In contrast, a computationally efficient and more practical approach to deriving sensitivity is to use the model's adjoint version, with the advantage of using only one forward and one backward simulation. In essence, one perturbs the target output and iterates the ocean model backward to determine the input conditions required to produce such a change. Since ocean dynamics encoded in the forward model propagate via the chain rule, one can trace the source along the backward integration. The gradient/sensitivity of the objective function to control variables reveals such vital information. Most importantly, the sensitivity of an objective function to the initial and boundary conditions and forcings can be estimated simultaneously (See schematic plot Fig. 2.1 b).

The diagram below (Figure2.2) illustrates the processes used to derive adjoint sensitivity in this investigation. First, a model is forced with surface forcings, such as surface wind stress and buoyancy fluxes, then integrated forward to simulate the global ocean circulation. Next, we define a scalar metric or objective/cost function characterizing the typical signal of large-scale, low-frequency sea level variation (temporally and spatially averaged sea surface height anomaly). The adjoint model, a transformation of the linearized forward model, is then integrated backward, calculating the gradients of the objective function to the control variables at various times and locations in model space. Consequently, the sensitivity to all key control factors can be evaluated simultaneously, which has the potential to discover the underlying processes that influence the objective quantity.

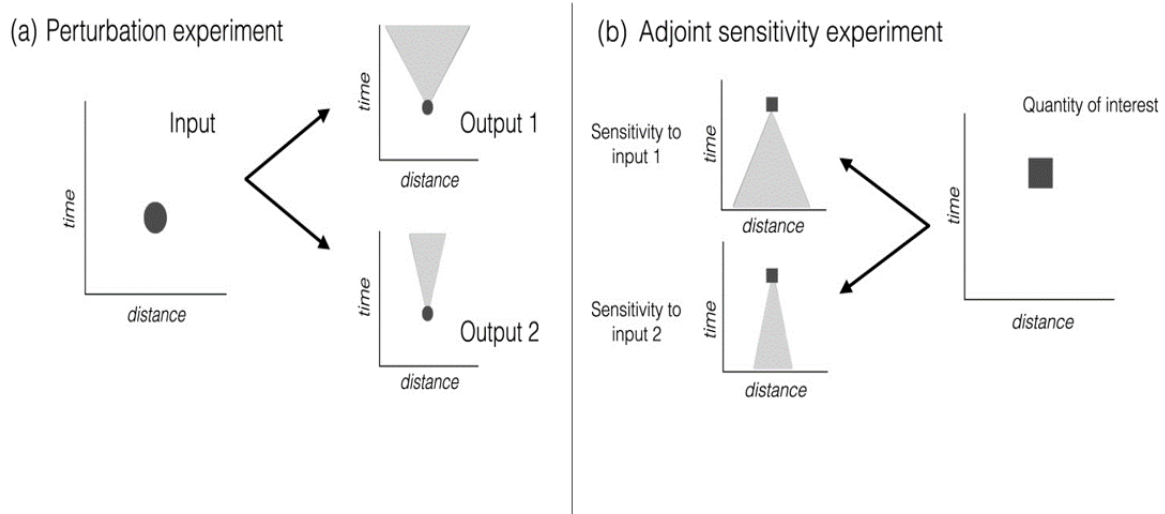


Figure 2.1: Schematic of a forward perturbation experiment (a), which produces various model outputs with one input perturbation. On the contrary, an adjoint sensitivity experiment (b) integrates backward compared with the forward model, which produces adjoint sensitivity to all control variables at all times and spaces. Figure taken from Jones et al. (2018)

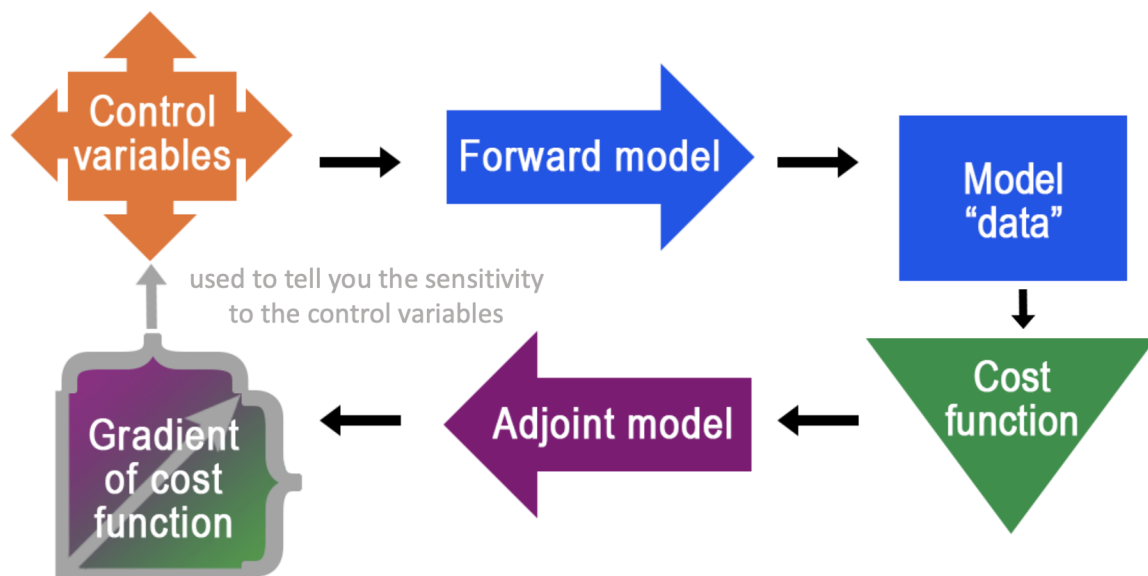


Figure 2.2: Schematic diagram of adjoint modeling used for sensitivity analysis adapted from the original diagram used to explain optimization problem. (<https://www.ecco-group.org/adjoint-overview.htm>)

The objective function design depends on the research's interest and should quantify the intrinsic character of the target quantity. The objective function can be generally categorized into two types: One type of objective function, as used in sensitivity studies, is a critical scalar quantity derived from model outputs, e.g., the present study's temporally and spatially averaged dynamical sea surface height anomaly. The gradients, or the sensitivity of the objective function to independent variables and previous model states, provide insight into the intricate mechanism (Marotzke et al., 1999). Another type is beyond the scope of this research but is extensively used in data assimilation. Using a least-squares method, the objective function can be the model-observation misfit, and the adjoint sensitivity can be used to adjust the initial conditions and estimate the best parameters, leading to a state of the ocean that is constrained by observations and consistent with model dynamics (Stammer et al., 1997, 2002).

One essential issue in deriving adjoint sensitivity is the availability of the adjoint model itself. The development of the adjoint model started with error-prone hand-coding in the early phase (Thacker and Long, 1988), which requires a comparable effort to develop the forward model. Once a tiny change is made to the forward model, the adjoint model needs to be rewritten. However, algorithmic differentiation (AD) tools have significantly boosted efficiency and avoided potential errors. Tangent linear and Adjoint Model Compiler (TAMC) and its commercial successor Transformation of Algorithms in Fortran (TAF) have been extensively used in the adjoint modeling (Giering and Kaminski, 1998). The software allows for fast computation and quantitative analysis of the gradients with exactness and completeness. Once the gradients of all control variables are estimated, they can be used in either sensitivity studies or optimization problems.

The principles of adjoint generation are well described in Giering and Kaminski (1998) and Marotzke et al. (1999). First, the compiler parses the forward model code to distinguish variables and syntax, followed by a data flow analysis to avoid trivial statements. Next, the adjoint code is constructed line by line based on mathematical parallelism, and finally, once a piece of adjoint code is constructed, the solution parts are determined. Interested readers are encouraged to read the literature for specific treatments and handling of conflicts.

Another feature of adjoint sensitivity worth mentioning is that it fundamentally differs from regression analysis. Adjoint sensitivity reveals statistical and causal relations encoded in the model integration between the independent variable and the target quantity. In contrast, correlation analysis describes how two quantities change with a similar pattern but do not necessarily prove causal connections (Zhang et al., 2011). Evaluating the adjoint sensitivity to each forcing provides insight into the separate mechanism. The statistical noise and correlation between different forcings can be avoided.

2.2 Mathematical formulation

Adjoint sensitivity can be explained explicitly from a mathematical perspective. Errico (1997) explicitly described adjoint sensitivity derivation using simple mathematics but used a least square objective function. Marotzke et al. (1999) extended this explanation by applying an automatic differentiation to produce the adjoint. Moore et al. (2009) adopted a propagator-based approach. Here, we review the mathematical derivation, keeping the essence but getting rid of complexity.

Consider a traditional nonlinear ocean model, the temporal evolution of the model state vector $\mathbf{x}$ can be represented as

$$\mathbf{x}_t = \mathcal{M}_t \cdot \mathcal{M}_{t-1} \cdot \mathcal{M}_{t-2} \dots \cdot \mathcal{M}_1 \mathcal{M}_0(\mathbf{x}_0) = \mathcal{M}(\mathbf{x}_0) \quad (2.1)$$

Where initial state and final state are unscribed by 0 and t , $\mathcal{M}$ stands for a nonlinear operator, a mapping from the input fields to output fields, or in other words, the temporal evolution of model states.

Typically but not absolutely, one wants to study a specific quantification of the oceanic states at the final time, which is usually a scalar metric called objective or cost function $\mathcal{J}$. The objective function is essentially a function of initial state $\mathbf{x}_0$ and thus may be regarded as model output for a more straightforward explanation.

$$\mathcal{J} = \mathcal{J}(\mathbf{x}_t) = \mathcal{J}(\mathcal{M}(\mathbf{x}_0)) \quad (2.2)$$

The cost function $\mathcal{J}$ can further be expressed via Taylor expansion with respect to a reference state.

$$\mathcal{J}(\mathbf{x}_t + \delta\mathbf{x}_t) = \mathcal{J}(\mathbf{x}_t) + \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T \delta\mathbf{x}_t + \mathcal{O}(\delta\mathbf{x}_t^2) \quad (2.3)$$

Note that the gradient is a co-vector. Therefore its transpose is required in the above inner product. The perturbation in the objective function is therefore approximated as

$$\delta\mathcal{J} = \mathcal{J}(\mathbf{x}_t + \delta\mathbf{x}_t) - \mathcal{J}(\mathbf{x}_t) = \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T \delta\mathbf{x}_t \quad (2.4)$$

A straightforward approach to calculate sensitivity is based on the forward perturbation experiments, which infers the difference in the objective function $\delta\mathcal{J}$ in response to a small perturbation in initial control variables $\delta\mathbf{x}_0$.

Linearizing the nonlinear model $\mathcal{M}$ about a reference trajectory, the evolution of the perturbed state can be represented as

$$\begin{aligned} \delta\mathbf{x}_t &= \mathbf{M}_t \cdot \mathbf{M}_{t-1} \cdot \mathbf{M}_{t-2} \dots \cdot \mathbf{M}_1 \cdot \mathbf{M}_0(\delta\mathbf{x}_0) \\ &= \mathbf{M}(\delta\mathbf{x}_0) \end{aligned} \quad (2.5)$$

with

$$M = \frac{\partial \mathcal{M}}{\partial \mathbf{x}_0} \quad (2.6)$$

Where M is the linearized approximation of the nonlinear operator $\mathcal{M}$ in terms of its Jacobian, which maps the input perturbations at the initial time into the output perturbations at the final time, it is referred to as the Tangent Linear Model (TLM).

The response in the objective function becomes,

$$\begin{aligned} \delta \mathcal{J} &= \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T M(\delta \mathbf{x}_0) \\ &= \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T M_t \cdot M_{t-1} \cdot M_{t-2} \dots \cdot M_1 \cdot M_0(\delta \mathbf{x}_0) \end{aligned} \quad (2.7)$$

Although TLM can readily be used to find the forward sensitivity, it quickly becomes prohibitive to proceed if there are many variables or the dimension is too large.

Alternatively, one can take advantage of the algebraic property of transpose,

$$\langle \vec{x}, A\vec{y} \rangle = \langle A^T \vec{x}, \vec{y} \rangle \quad (2.8)$$

Thus, the response in the objective function is:

$$\begin{aligned} \delta \mathcal{J} &= \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T M(\delta \mathbf{x}_0) \\ &= M^T \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T (\delta \mathbf{x}_0) \\ &= M_0^T \cdot M_1^T \dots \cdot M_{t-2}^T \cdot M_{t-1}^T \cdot M_t^T \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \right]^T (\delta \mathbf{x}_0) \\ &= \left[\frac{\partial \mathcal{J}}{\partial \mathbf{x}_0} \right]^T (\delta \mathbf{x}_0) \end{aligned} \quad (2.9)$$

Where M^T is the transpose of TLM, known as the adjoint operator, which is composed of continuous backward products via the chain rule, mapping the “sensitivity gradient” backward in time to the initial time step and yielding the sensitivity gradient with respect to x_0 , referred to as $(\partial J / \partial x)$:

$$\frac{\partial \mathcal{J}}{\partial \mathbf{x}_0} = M^T \frac{\partial \mathcal{J}}{\partial \mathbf{x}_t} \quad (2.10)$$

In contrast to the forward sensitivity, adjoint sensitivity propagates backward in time with one simulation to trace the influences from all controls simultaneously.

2.3 Application in oceanography

The adjoint-based sensitivity reveals causal relationships between objective quantity and model variables because the physics is propagated through the chain rule in the model. Therefore, adjoint sensitivity has been extensively used to study the causal mechanism of oceanic quantities.

Marotzke et al. (1999) pioneered an adjoint sensitivity analysis using the ocean general circulation model to study the dynamical causes of Atlantic meridional heat transport. Since then, the underlying driving mechanism of Atlantic meridional overturning circulation has been extensively investigated (Köhl, 2005; Bugnion et al., 2006b; Czeschel et al., 2010; Pillar et al., 2016; Kostov et al., 2019; Smith and Heimbach, 2019). Besides, the application of adjoint sensitivity has been extended to a variety of interests, including ENSO (Zhang et al., 2011), biogeochemistry (Dutkiewicz et al., 2006, 2009), ice-shelf (Heimbach and Losch, 2012), ocean current (Czeschel et al., 2012; Gopalakrishnan et al., 2013; Zhang et al., 2011), ocean heat content (Jones et al., 2018), ocean bottom pressure (Fukumori et al., 2015), eddies (Zhan et al., 2018), etc.

In terms of studying sea level using the adjoint method, the driven mechanism or causation of regional sea level variability can be revealed based on adjoint gradient decomposition and reconstruction. Specifically, the sensitivity is convolved with forcing anomalies to reconstruct sea level variation and attribute variability to individual forcing. To name a few publications, Fukumori et al. (2007, 2015, 2021) extensively investigated regional sea level variability in the Mediterranean sea, Arctic Ocean, and Nordic Seas, illustrating the use of the adjoint sensitivity to identify causal mechanisms. Verdy et al. (2014) studied the physical pathways of wind-driven influences on sea level variability on the California coast. Wang et al. (2022) attributed the influences of local and remote forcing on interannual sea-level variability at Nantucket Island. Frederikse et al. (2022) examined the seasonal predictability of sea level anomaly in the United States east coast by convolving adjoint sensitivity with a concatenated forcing anomaly.

In addition to sensitivity studies, adjoint sensitivity can be implemented in data assimilation, parameter estimation, stability analysis, etc. (Errico, 1997). Taking account of massive amounts of observation, the objective function is typically selected to be a scalar-valued model-data misfit, synthesizing information gained from observation and model simulation. The adjoint-based sensitivity can reduce spurious model drifts and biases through the inversion of uncertain parameters, rendering an observation-constrained, physically consistent state estimation (Forget et al., 2015). Such state estimate is an essential component of the Consortium for Estimating the Circulation and Climate of the Ocean (ECCO) project, which combines cutting-edge ocean models with observational data sets (See Fig 2.3) to produce the most accurate estimates of ocean circulation and its influence in climate (Wunsch et al., 2009; Heimbach et al., 2019).

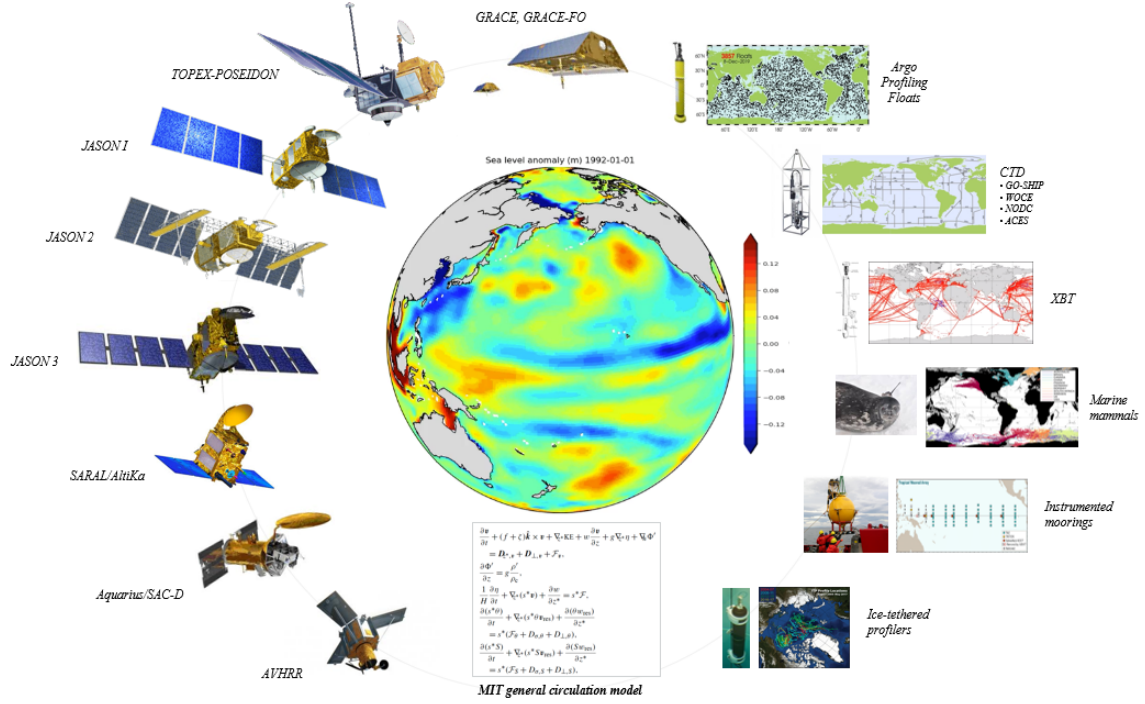


Figure 2.3: ECCO state estimates are multi-platform, multi-instrument synthesis products that integrate ocean and ice observations and models. (Fenty, 2020)

The more advanced applications include Hessian-based uncertainty quantification in inverse problems and Hessian-based optimal experimental design Kalmikov and Heimbach (2014); Moore et al. (2018). Furthermore, the additional value of the adjoint sensitivity is the observation systems design (Köhl and Stammer, 2004).

Even though adjoint sensitivity has led to significant findings in numerous domains, many intriguing questions still require further investigation. The limitations will be elaborated on in the latter chapter, using this study as an example.

2.4 EOF mode

Previous research has highlighted the Indian sector of the Southern Ocean as the region of interest due to a barotropic mode. Here, we employ an empirical orthogonal function (EOF) analysis to examine the spatial scale of variability using the outputs of one hundred MPI-GE members.

The first EOF mode exhibits enhanced SSH variability, centered off the Southwest of Australia, with a core largely confined in $[48-56^\circ\text{S}, 88-120^\circ\text{E}]$ for most members (See Fig 2.4). The pattern is strikingly similar to that in (Webb and de Cuevas, 2002).

A 36-month lowpass filter is applied to determine the low-frequency mode's spatial extent. As a result, the first EOF mode is still largely present in the region of the high-frequency mode but to a considerably lesser extent.

Note that the first EOF mode covers the same region in S.D. Nandini-Weiss et al. (submitted to J. Geophys. Res. Ocn), which they found a significant change in sea level variability under the global warming scenario. Therefore, we chose the core of the first EOF mode (applied 36-month lowpass filter) as the region of interest in this study, which exhibits heightened low-frequency sea level variability ($[52-56^\circ, 96-112^\circ\text{E}]$, See Fig 2.5, the red box).

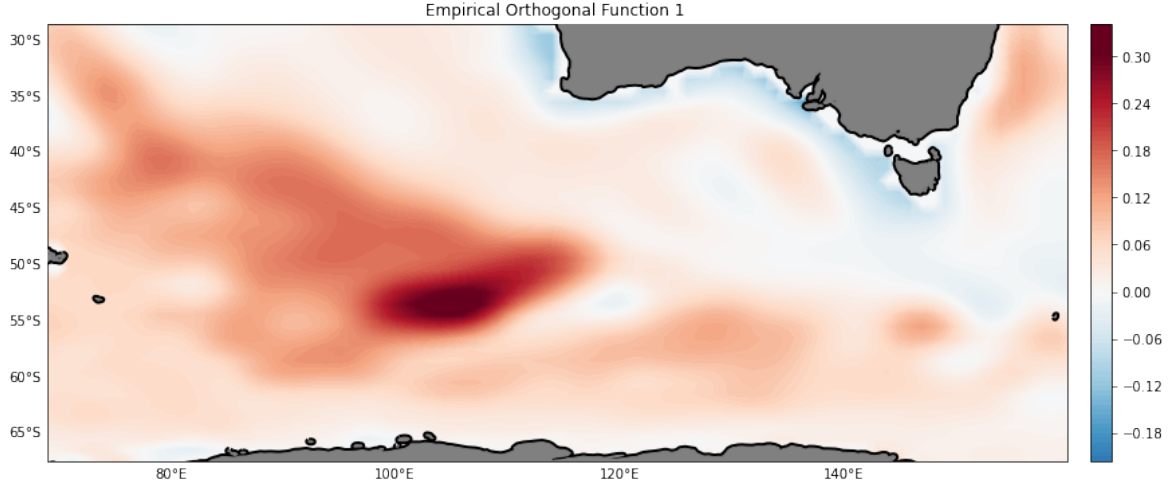


Figure 2.4: First EOF mode of SSH anomalies for the first 100 years of historical scenario based on one member output of MPI-GE.

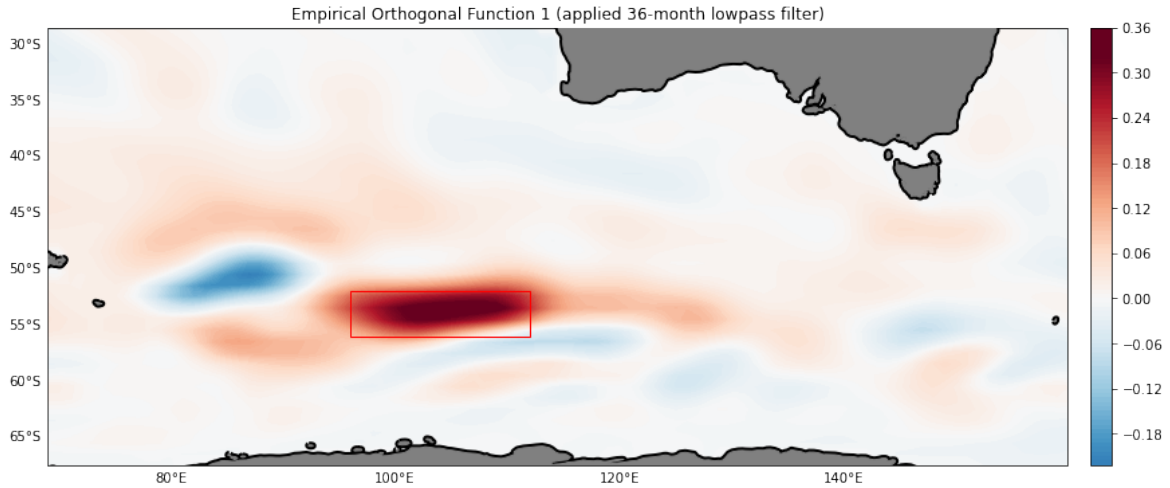


Figure 2.5: Same with the previous plot, but applied a 36-month lowpass filter before EOF analysis. The red box ($[52-56^\circ\text{S}, 96-112^\circ\text{E}]$) indicates the region of interest in this study.

Chapter 3

Methodology

In the present study, the Massachusetts Institute of Technology General Circulation Model (MITgcm) and its adjoint model are exploited to investigate the sensitivity of regional sea level variability to atmospheric forcing and previous states. In this chapter, we describe the configuration of both the forward and the adjoint model, the design of the objective function for the adjoint model, and sensitivity-based forward perturbation experiments.

3.1 MITgcm and its adjoint

The Massachusetts Institute of Technology general circulation model (MITgcm, see online at <http://mitgcm.org/>) is a state-of-the-art ocean model which solves Navier-Stokes equations using the finite volume method (Marshall et al., 1997a,b). The model employs Boussinesq approximation for an incompressible fluid on discretized curvilinear grids (Arakawa and Lamb, 1977). Its flexible non-hydrostatic formulation enables efficient simulation for fluid phenomena ranging from small-scale convection to large-scale global circulation.

In particular, MITgcm has been adapted to generate the tangent linear and adjoint model along with the forward model, enabling sensitivity study, parameter optimization, and state estimate (Marotzke et al., 1999; Stammer et al., 1997, 2002). The adjoint capability is essential for sensitivity analysis in the present study.

The tangent linear and adjoint model can be obtained automatically with algorithmic/automatic differentiation (AD) tools, which apply the chain rule to compute the first derivative of an objective function to the control variables, functioning as a source-to-source translator. In this study, the commercial software TAF (Transformation of Algorithms in Fortran, see online at <http://fastopt.de/>) is utilized to build the adjoint model automatically.

3.2 Forward model

The forward model configuration modulated from the MITgcm tutorial experiment “global_ocean.90x40x15” simulates the planetary ocean circulation. The similar configuration has been repeatedly tested and justified in various literature serving different purposes (Hill et al., 2004; Losch et al., 2004; Ferreira et al., 2005; Bugnion et al., 2006a,b; Losch and Heimbach, 2007; Allison et al., 2010). The quasi-global domain covers from 80 °S to 80 °N, with a horizontal resolution of 4°x4° spherical polar grid and fifteen vertical layers with thicknesses ranging from 50 meters near the surface to 690 meters at the bottom, giving a maximum model depth of 5200 meters. The topography is derived from the Earth Topography Five-Minute Grid (ETOPO5, see online at <https://www.ngdc.noaa.gov/mgg/global/etopo5.HTML0>) based on the satellite (Smith and Sandwell, 1997), with smoothing, averaging and interpolation to the 4° grid (see Fig 3.1).

The forward model is forced by climatological monthly mean zonal and meridional wind stress fields (Taux, Tauny) from Trenberth et al. (1990), surface net heat flux (Qnet), and surface net freshwater flux (Empr, stands for evaporation, precipitation, and runoff) from NCEP/NCAR re-analysis (Kalnay et al., 1996). These forcings are repeated every 12 months. Hence the model has no externally forced interannual variability. Additionally, sea surface temperature and salinity are relaxed to climatological values from the World Ocean Atlas (Levitus and Boyer, 1994; Levitus et al., 1994) on a timescale of two-month and six-month period, respectively, to reduce model drift. Note that we do not include additional parameterizations (e.g., Gent-McWilliams and Redi’s eddy parameterization), which could complicate our efforts to study large-scale surface circulation.

Following a 100-year spin-up integration, the forward model is integrated for ten years to avoid problems with initial adjustment processes. The forward integration determines the background state, which the adjoint model is linearized around.

The model adopts a hydrostatic configuration with an implicit free surface. The timesteps used for the momentum equations are 20 minutes, and for tracer and implicit free surface equations are 12 hours. An explicit configuration of the critical parameters is listed in table 3.1.

Table 3.1: **Model parameters**

Parameters	Model symbol	Value	Unit
vertical dissipation coefficient	viscAr	1×10^{-3}	$m^2 s^{-1}$
horizontal frictional dissipation coefficient	viscAh	5×10^5	$m^2 s^{-1}$
horizontal diffusion coefficient for T/S	diffKhT/diffKhS	0	
vertical diffusion coefficient for T/S	diffKrT/diffKrS	3×10^{-5}	$m^2 s^{-1}$
time step for momentum	deltaTmom	1200	s
default time step	deltaTClock	43200	s

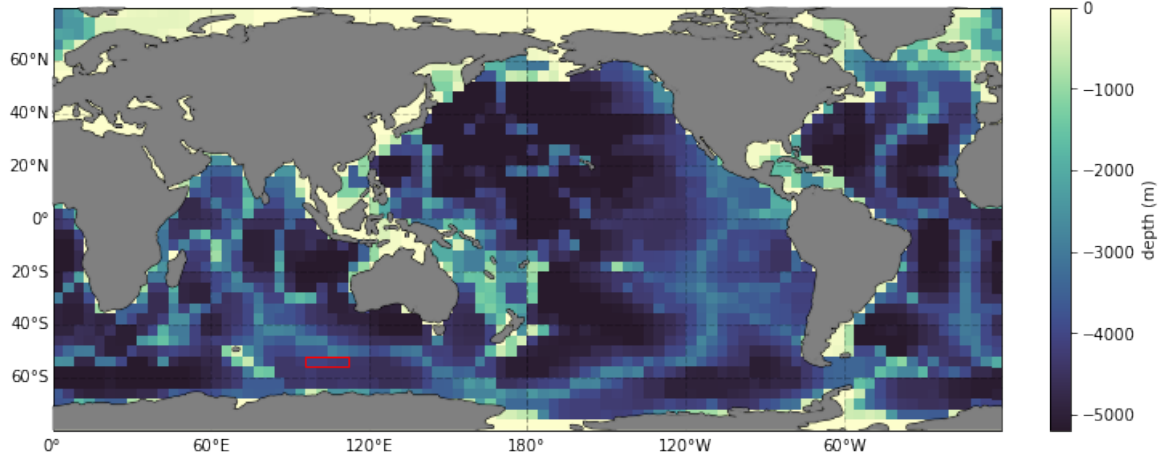


Figure 3.1: Model bathymetry, the red box indicates the mode region, where the objective function is evaluated.

A set of equations represented in the model are solved as follows, where zonal and meridional wind stress forcing ($\mathcal{F}_u, \mathcal{F}_v$) are added to the momentum equations in (3.1) and (3.2). Thermodynamic forcing inputs ($\mathcal{F}_\theta, \mathcal{F}_S$) calculated from surface heat flux and surface freshwater flux are represented in thermodynamics equations for potential temperature θ (3.4) and salinity S (3.5).

$$\frac{Du}{Dt} - fv + \frac{1}{\rho} \frac{\partial p'}{\partial x} - \nabla_h \cdot (A_h \nabla_h u) - \frac{\partial}{\partial z} \left(A_z \frac{\partial u}{\partial z} \right) = \begin{cases} \mathcal{F}_u & \text{(surface)} \\ 0 & \text{(interior)} \end{cases} \quad (3.1)$$

$$\frac{Dv}{Dt} + fu + \frac{1}{\rho} \frac{\partial p'}{\partial y} - \nabla_h \cdot (A_h \nabla_h v) - \frac{\partial}{\partial z} \left(A_z \frac{\partial v}{\partial z} \right) = \begin{cases} \mathcal{F}_v & \text{(surface)} \\ 0 & \text{(interior)} \end{cases} \quad (3.2)$$

$$\frac{\partial \eta}{\partial t} + \nabla_h \cdot \vec{u} = 0 \quad (3.3)$$

$$\frac{D\theta}{Dt} - \nabla_h \cdot (K_h \nabla_h \theta) - \frac{\partial}{\partial z} \left(\Gamma(K_z) \frac{\partial \theta}{\partial z} \right) = \begin{cases} \mathcal{F}_\theta & \text{(surface)} \\ 0 & \text{(interior)} \end{cases} \quad (3.4)$$

$$\frac{DS}{Dt} - \nabla_h \cdot (K_h \nabla_h S) - \frac{\partial}{\partial z} \left(\Gamma(K_z) \frac{\partial S}{\partial z} \right) = \begin{cases} \mathcal{F}_S & \text{(surface)} \\ 0 & \text{(interior)} \end{cases} \quad (3.5)$$

$$g\rho_0\eta + \int_{-z}^0 \rho' dz = p' \quad (3.6)$$

The above Navier–Stokes equation set represents the core physics contained in the model, where u and v are the zonal and meridional components of the flow vector

$\bar{\mathbf{u}}$, f is the Coriolis parameter. For other parameters, readers may refer to the MITgcm manual (https://mitgcm.readthedocs.io/en/latest/examples/global_oce_latlon/global_oce_latlon.html).

As a model with a coarse resolution, the current setup does not resolve mesoscale or microscale activities but provides a smooth and simplified background field. Although small-scale sensitivities within a $4^\circ \times 4^\circ$ grid are suppressed, the configuration highlights the large-scale patterns, and previous similar configurations imply that the model is capable of reproducing large-scale processes and their variability. More importantly, it is an inexpensive and adequate setup that can be implemented for thousands of years on a desktop computer with a single processor.

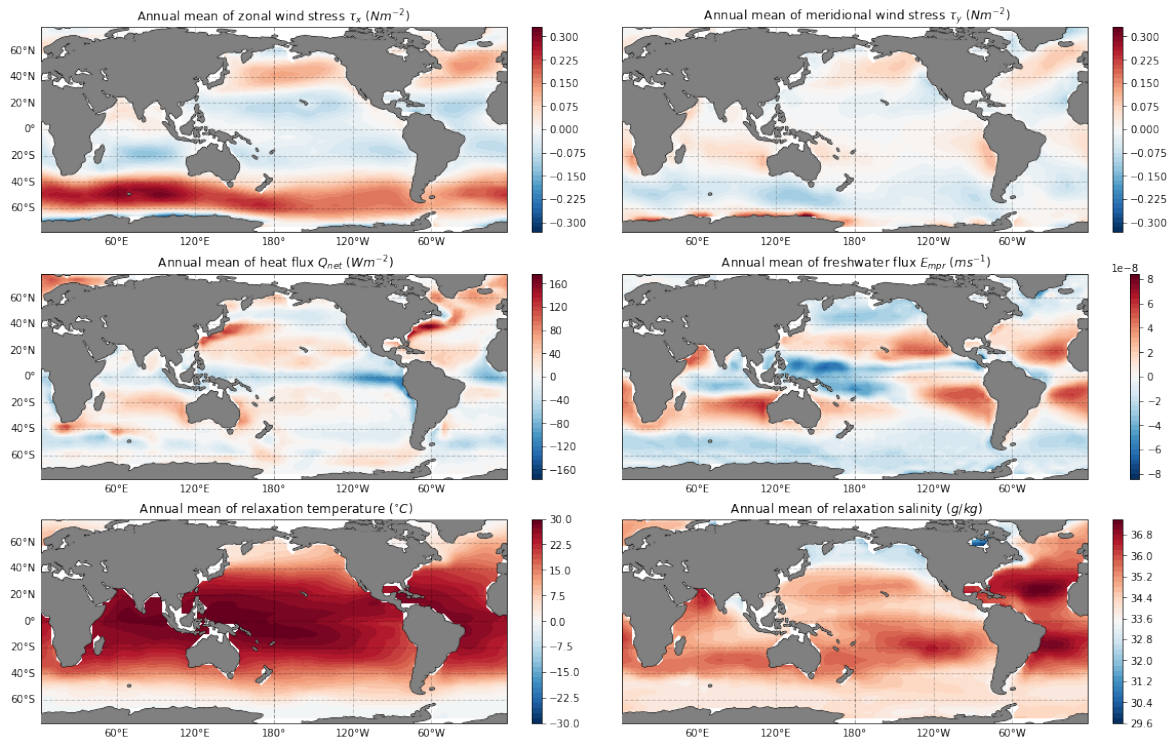


Figure 3.2: Model inputs, plotted as annual mean.

3.3 Adjoint model

The adjoint model inherits all the configurations from the forward model. The adjoint model of MITgcm is built using the automatic differential tool, the Transformation of Algorithms in the Fortran compiler (TAF) (Giering and Kaminski, 1998), which applies the chain rule to the forward model code to automatically generate the tangent linear model and adjoint model line by line.

In this work, the model does not simulate GMSL explicitly. Instead, we investigate dynamical sea level η , which reveals regional deviation of sea level compared with the

global mean. It is a crucial indication of changes in the upper ocean circulation and horizontal density gradient.

To characterize the low-frequency sea-level variability in the box region, we construct an objective function J , quantifying the three-year mean sea level height anomaly averaged over a box area.

$$J = \frac{1}{(t_2 - t_1) A} \int_{t_1}^{t_2} \int_A \eta \, dA \, dt \quad (3.7)$$

Where η represents the dynamic sea surface height anomaly, A is the box region covering the maximized signals in the first mode of EOF analysis (52-56°S, 96-112°E, see Fig 2.5)), t stands for a 10-years adjoint run, and $t_1 - t_2$ covers the last three years for the evaluation of the objective function. The objective function J is a single-value metric with a unit of meter, which is evaluated as a function of ocean states from the forward solution. With a time duration of three years and a spatial size of the box, the objective function quantifies the large-scale, low-frequency fluctuation of sea levels in this region.

The adjoint sensitivity of the objective function J to the control variables x , including initial conditions, boundary conditions, state variables, model parameters, and input forcings, is expressed as:

$$\frac{\partial J}{\partial x}(\mathbf{r}, t) \quad (3.8)$$

Where x can be any model parameters, forcing fields, or state variables, t is the time dimension, $\mathbf{r}$ represents the two-dimensional space if x is surface forcing, and can be three-dimensional if x is three-dimensional oceanic state fields. Note that this study focuses on the atmospheric forcings, namely the surface wind stress, surface net heat and freshwater flux. Therefore, the sensitivity to the surface forcings is three-dimensional and time-varying.

The obtained sensitivity has several applications:

- Identify dominant sensitivity pathways, tempo-spatial scales, and associated dynamical processes that can potentially affect the objective quantity.
- The sensitivity may be helpful to for oceanographic observation design,
- As a foundation for subsequent data assimilation or optimization problem.

Over the ten years of integration, sensitivity evolves slowly. Therefore snapshots of the adjoint sensitivity fields are saved at monthly intervals.

3.4 Numerical experiments

Note that the adjoint-derived sensitivity is the linear approximation of the nonlinear equations. The linear approximation is valid when the perturbation is small and the integration time is short. A set of nonlinear forward perturbation experiments are conducted.

Such perturbation experiments serve two purposes: (1) verify the linear assumption of the adjoint method by comparing the perturbed model trajectory with the reference run. (2) highlight the physical processes induced by the forcing perturbation.

Since we focus on large-scale and low-frequency variability rather than specific events, the forward model is driven by climatology. Therefore, the forward solution does not necessarily have to be fitted or compared with observation. We assume that the sensitivity is independent of the state of the ocean, considering the background states are in equilibrium (Heimbach et al., 2011; Zhang et al., 2011). Furthermore, we validated that the sensitivity pattern hardly changes when the forward model is driven by forcings representing RCP scenarios; therefore, we conclude that the objective function determines the adjoint sensitivity in an equilibrium state.

For the sake of clarity, all numerical experiments done in this work are listed below:

- The ten-year integration of a forward model establishes the baseline state. The forward solution can demonstrate its capability in simulating large-scale circulation by checking some metrics. In addition, it serves as a reference for comparison with perturbation experiments.
- An adjoint model is integrated backward, calculating the first derivative (sensitivity) of the objective function to independent variables. Note that the adjoint model configuration inherits the forward model.
- A set of wind perturbation experiments with sensitivity-like wind stress perturbations, which can highlight relevant physical processes contained in the model and examine the validity of the linearity assumption.
- A heat flux perturbation experiment, highlighting the physical processes associated with buoyancy components.

Chapter 4

Results

In this chapter, we begin by introducing several crucial aspects that illuminate the forward solution's states. Next, we describe the spatial patterns and temporal evolution of the adjoint sensitivity and associated physical processes. Then, we illustrate the scaled sensitivity, which has the same units for different atmospheric forcing variables. It is obtained by multiplying the sensitivity by its respective forcing standard deviations under the global warming scenario. Thus, we can quantify the relative contribution of each surface forcing in a specific condition. Finally, we employ forward perturbation experiments to distinguish between local and nonlocal wind impacts and contrast the differences between wind stress and buoyancy components.

4.1 Forward solution

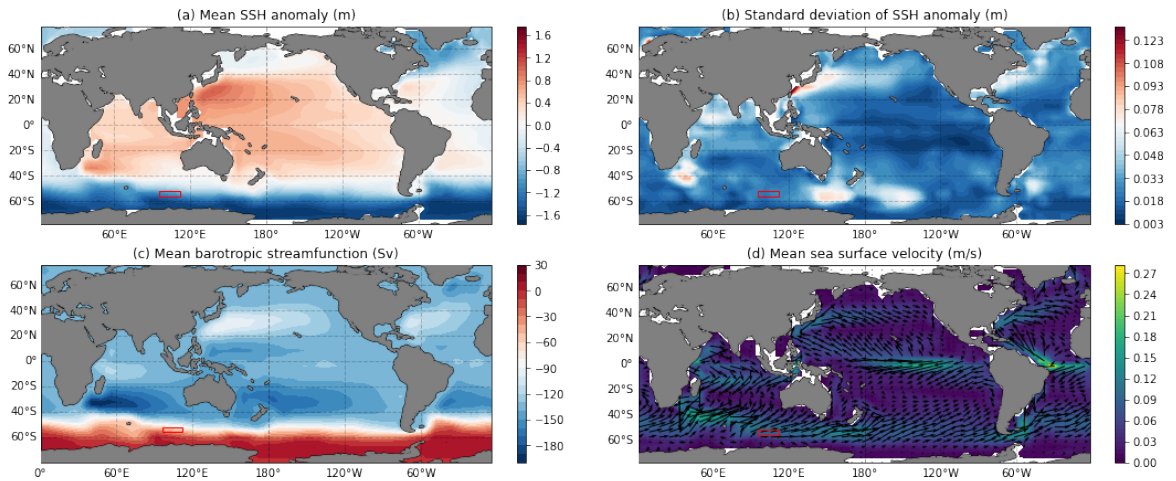


Figure 4.1: The mean states result from a ten-year forward integration. (a) mean SSH anomaly, (b) standard deviation of SSH anomaly (c) mean barotropic streamfunction, and (d) mean sea surface velocity. The red box indicates the region of interest.

After a ten-year integration, the model reaches an equilibrium state. Figure 4.1 depicts the mean states of the global ocean over ten years:

- (a) The forward model simulates the dynamic sea level anomaly, which is positive in the low latitudes and negative in the high latitudes. It is characterized by a strong north-south gradient in the Southern Ocean and the most elevated SSH in the Western Pacific due to ocean circulation.
- (b) The standard deviation of monthly SSH anomalies reveals regional interannual ocean variability in the major ocean current systems, highlighting regions that are more variable than others. These regions are primarily the western boundary systems, such as the Kuroshio Current near Japan, the Gulf Stream in the North Atlantic, and the Agulhas Current in the Indian Ocean. Furthermore, the ACC also demonstrates strong variability, especially in the Pacific sector.
- (c) The mean barotropic streamfunction summarizes the vector flow field quantitatively. It shows major subtropical gyres in ocean basins and the ACC transport, indicating that the forward model can simulate critical features of a realistic ocean. However, the simulated ACC has a much weaker strength, which is less than a reasonable estimate. It might be due to the coarse resolution and parameterizations we chose not to use.
- (d) The mean sea surface velocity presents surface currents around the globe, such as the western boundary and equatorial currents. The ACC current shows a northeast deflection, with a mean velocity of 0.2-0.25 m/s at the surface.

Despite some fine features being absent due to the model coarse resolution, the simulated mean SSH anomaly and its variability are reasonable and consistent with the satellite observation (e.g., Archiving, Validation and Interpretation of Satellite Oceanographic data, AVISO. <https://www.aviso.altimetry.fr/en/applications/ocean/large-scale-circulation/mean-dynamic-topography.html>).

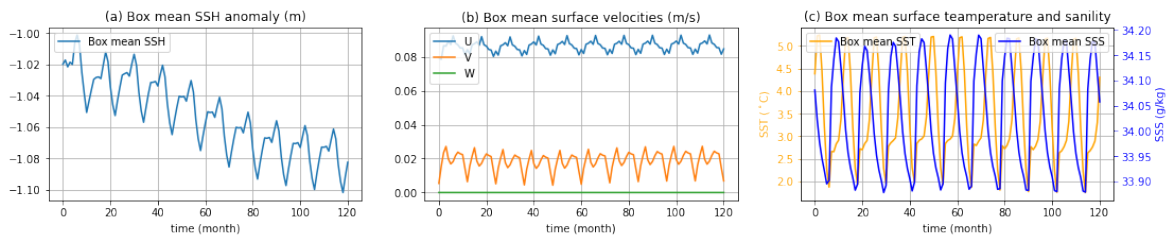


Figure 4.2: Time series of key oceanic variables. (a) mean SSH anomaly averaged over the box region, (b) three components of surface velocities in the first layer averaged the box region, and (c) box mean SST and SSS at the surface.

Figure 4.2 shows the time series of critical prognostic variables in the surface layer within the target box, where the objective function is defined:

- (a) The spatial mean SSH anomalies in the target region remain negative and decrease slightly resulting from the initial adjustment, with a significant annual cycle.
- (b) The spatial mean horizontal surface velocities in the target region are positive, with yearly cycles. The zonal velocity is the most substantial, while the vertical velocity is negligible.
- (c) Similarly, the averaged sea surface temperature and salinity oscillate with annual cycles.

These time series do not reveal the internal variability in the box region, as driven by the monthly climatology forcing. The ocean in the region of interest is essentially in equilibrium during the ten years.

4.2 Sensitivity pathways and processes

Note that the objective function is the last 3-year mean SSH anomaly averaged over a 1x4 grid box region (52-56° S, 96-112° E), and the adjoint model is integrated ten years backward in time. Due to the intrinsic feature of backpropagation, we describe the sensitivities in reverse chronological order. The end time step of the forward model, which is the initial time step of the adjoint model, is labeled as time 0 and goes backward until year -10. We examine the monthly snapshots of sensitivity fields at various lead times and associated physical processes at the surface ocean. The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal at each time frame.

The adjoint model calculates the linear sensitivity (first derivative) of the objective function with respect to control variables, including zonal surface wind stress ($\partial J/\partial \tau_x$), meridional surface wind stress ($\partial J/\partial \tau_y$), net surface heat flux from the ocean to the atmosphere ($\partial J/\partial Q_{net}$), net surface fresh water flux from the ocean to the atmosphere ($\partial J/\partial Empr$), potential temperature ($\partial J/\partial T$), salinity ($\partial J/\partial S$), sea surface height anomaly ($\partial J/\partial \eta$), three components of velocity fields ($\partial J/\partial U, \partial J/\partial V, \partial J/\partial W$) or other model controls. We are interested in how a perturbation in external forcing affects the variability of the objective function, so we focus our analysis on surface zonal and meridional wind stress, net heat flux, and freshwater fluxes.

Adjoint sensitivity can shed light on the regional sea level variability mechanism by examining spatial patterns and temporal evolution of sensitivity fields, which reveal critical physical processes. The interpretation of adjoint sensitivity is that a positive/negative sensitivity value at a location indicates that the value of the objective function will increase/decrease by the given magnitude if a positive perturbation is imposed on the respective forcing over the grid cell for one month. The influence of

forcings on the objective function at various tempo-spatial scales can be estimated qualitatively by looking into sensitivity propagation. Their relative contribution under the global warming scenario can be evaluated quantitatively by comparing the scaled sensitivity in response to forcing uncertainty.

4.2.1 Sensitivity to the wind stress

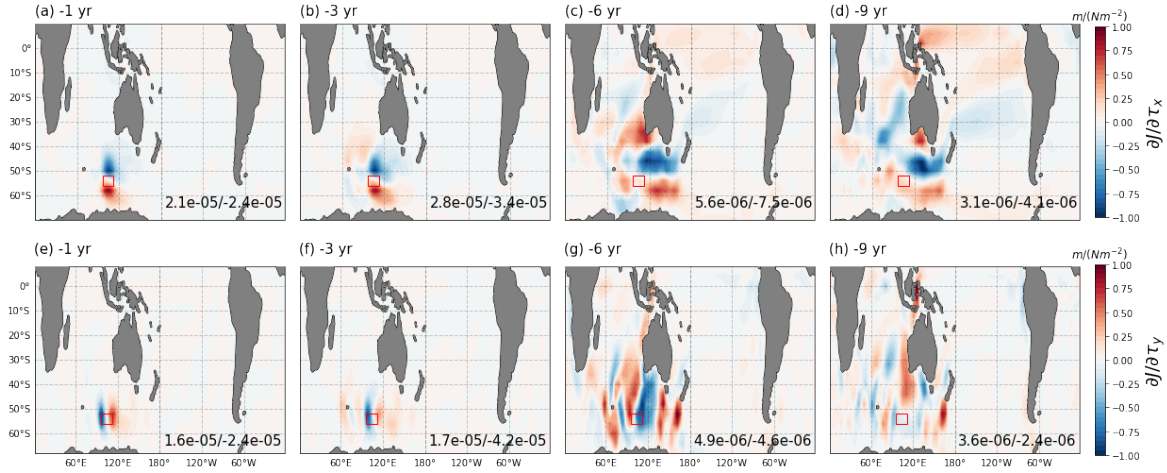


Figure 4.3: Evolution of the adjoint sensitivity (from left to right, backward in time) with respect to zonal wind stress ($\partial J/\partial\tau_x$, subplots a-d) and meridional wind stress ($\partial J/\partial\tau_y$, subplots e-h). The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative sensitivity in each snapshot. The red box indicates the region where the objective function is defined.

Figure 4.3 shows the sensitivities to zonal surface wind stress ($\partial J/\partial\tau_x$, subplots a-d) and meridional surface wind stress ($\partial J/\partial\tau_y$, subplots e-h) at various time lags. The sensitivities to the wind components are confined mainly to the southern hemisphere over the ten years. The dominant signals are assembled in the Southern Ocean Indian Sector, indicating influential local and regional processes. However, there are obvious signals in the equatorial and mid-latitude oceans, especially at longer time lags. The teleconnection reflects a linkage between the circulation of the Pacific and Indian Oceans via oceanic bridges, such as the Indonesian throughflow.

The maximum and minimum sensitivities in the corresponding time are labeled in each subplot. The magnitude of sensitivities to wind stress components increases from the beginning and reaches a maximum at around -3 yr, over which the sensitivities accumulate. After that, the sensitivities gradually decay backward in time.

At -1 year, the sensitivities to the surface wind stress encompass the box. The sensitivities to the zonal wind stress spread north and south of the box symmetrically. Likewise, the sensitivities to the meridional wind stress lay west and east. For $\partial J/\partial\tau_x$,

the positive maximum locates at the south of the box while the negative minimum locates at the north of the box, exhibiting a dipole pattern. It is similar for $\partial J/\partial\tau_y$, the positive maximum locates at the east, and the negative minimum finds at the west of the box. The symmetrically distributed patterns form an anti-cyclonic wind vortex in the Southern hemisphere, indicating a wind stress curl around the target region, which can evoke Ekman pumping in the upper ocean.

At -3 years, the primary structures are unchanged, but more sensitivities start to emerge. Take zonal wind stress, for example. Sensitivities appear on the west side of the dipole with adverse signs. A red patch is located off the southwest of Australia, and a vague blue patch shows up right below the red patch. Positive sensitivities alternate with negative sensitivities, exhibiting a symmetry pattern. These dipole-type structures may be associated with Ekman pumping and suction (Czeschel et al., 2012; Jones et al., 2018).

At -6 years, the sensitivities to the wind stress spread to a broader region. For $\partial J/\partial\tau_x$, the dipole patterns are shifting eastward. Meanwhile, the sensitivities in the remote regions are becoming more noticeable but still less significant than the local signals. The sensitivities at the equator suggest Kelvin waves. The sensitivities in the subtropical gyres may be associated with planetary Rossby waves. Those signals exhibit a southwest-to-northeast tilt, a signature of the β effect, reflecting the wave speed proportional to $\beta = \partial f/\partial y$. For $\partial J/\partial\tau_y$, the sensitivities show a north-south extension. The sensitivities through the Indonesian Passage and the northwest coast of Australia become evident, implying a connection between the Indian and Pacific Ocean modulated by the meridional wind stress.

At -9 years, the sensitivity patterns resemble stripes, suggesting wave propagation. The dipole remains quasi-stationary between Australia and Antarctica, while sensitivities in the Indian and Pacific basins shift eastward. Sensitivities to zonal wind stress in the Indian Ocean have a pattern almost the same as sensitivities in the Pacific basin, suggesting similar planetary waves in the ocean basins. The equatorial sensitivities become visible, indicating the teleconnection of equatorial Kelvin wave. Sensitivities to meridional wind stress are prominent along Australia's west and southeast coasts, Indonesian seas, and the Peru coast, reflecting coastally trapped or topography-influenced waves modulated by meridional wind stress.

The sensitivities to the surface wind stress components reveal the optimal temporal-spatial wind stress pattern, in which the surface wind stress can affect the regional SSH variability. On the assumption that these optimal wind at work, as implied by the sensitivities, the pattern of the surface wind stress curl ($\partial\tau_y/\partial x - \partial\tau_x/\partial y$) can be derived according to the formula $\left[\frac{\partial J/\partial\tau_y}{\partial x} - \frac{\partial J/\partial\tau_x}{\partial y}\right]$. An anti-cyclonic horizontal wind stress shear indicated by wind stress sensitivities corresponds to a curl pattern resembling sensitivity to surface wind stress curl. The large-scale wind stress curl indicates Ekman convergence or divergence at the ocean surface that leads to interior downwelling or upwelling.

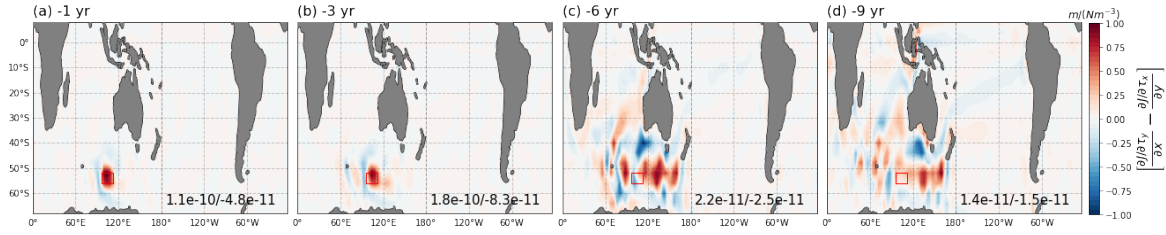


Figure 4.4: Evolution of the optimal surface wind stress curl (from left to right, backward in time), derived from the sensitivities to the surface wind stress components. The values are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative values in each snapshot. The red box indicates the region where the objective function is defined.

Figure 4.4 shows the optimal wind stress curl pattern derived from the sensitivities to the surface wind stress components. Positive curl indicates ocean convergence and downward Ekman pumping, and negative curl implies ocean divergence and Ekman upwelling in the Southern Hemisphere.

At -1 year and -3 years, it shows a dominant positive surface wind stress curl around the box region and opposing surface wind stress curls surrounding the positive curl, demonstrating the primary Ekman convergence and downwelling near the box and Ekman upwelling outside. Specifically, an intense positive wind stress curl over the box drives Ekman convergence on the surface ocean, promoting a positive SSH anomaly and Ekman downwelling in the interior water column. Meanwhile, the outside of the box exhibits negative sensitivity, indicating Ekman upwelling processes driven by negative wind stress curl.

At -6 years, the curl pattern becomes more complex, occupying a larger region. The dominant positive wind stress curl has shifted east along the ACC zonal band. Negative wind stress curls are found north and south of positive curls. In addition, there are tilted, stripe-like curls in the Indian and Pacific oceans, suggesting large-scale waves influenced by surface forcing.

At -9 years, the leading positive wind stress curl is still in the east of the box, between Australia and Antarctica. There are also less significant wind stress curls in the Indonesian Passage and open oceans, indicating Indonesian throughflow and planetary waves.

In a forward sense (from -10 to 0 years), the optimal positive curl forms in the southwest of New Zealand and propagates westward. Similarly, the primary negative signal comes from the south coast of Australia and also propagates westward along the positive curls.

We further check the sensitivity to the SSH anomaly to verify if wave propagation exists between Australia and Antarctica. As a result, the sensitivity to the SSH anomaly is positive in the whole domain, and we look into its anomaly compared with its global mean.

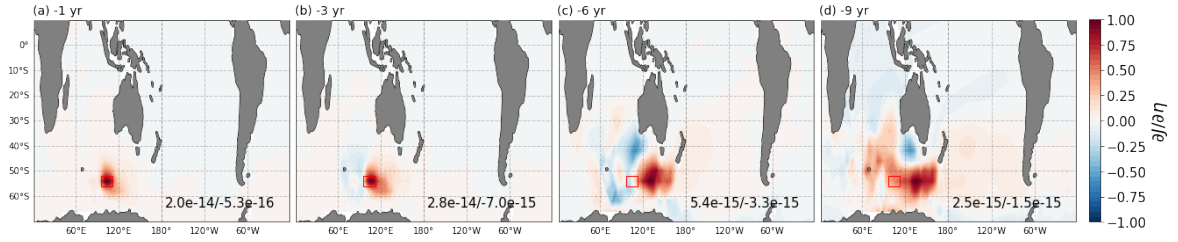


Figure 4.5: Evolution of adjoint sensitivities (from left to right, backward in time) with respect to the SSH anomaly ($\partial J/\partial \eta$). The unit has been canceled, as J and η have the same unit. The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative sensitivity. The red box indicates the region where the objective function is defined.

Figure 4.5 shows the evolution of sensitivities to SSH anomaly. The sensitivity pattern simply reflects the spatial scale and temporal propagation of SSH anomaly. In a forward sense (from -10 to 0 years), the optimal positive SSH anomalies originate from the east of the box and move westward, with sea level depression accompanied by the surroundings. It suggests a westward-propagating wave in the Southern Ocean and is likely a result of the doppler-shifted westward-propagating Rossby wave advected by the eastward-moving ACC (Heimbach et al., 2011).

A Hovmöller diagram of SSH sensitivity anomaly averaged in the ACC sector (42-62°S) clearly shows the westward propagation of SSH signals (Fig 4.6), indicating the location and timing of the wave pathway. A crude phase speed estimation by calculating its slope is 1.88 cm/s , comparable to the theoretical Rossby wave phase speed estimated in (Chelton and Schlax, 1996).

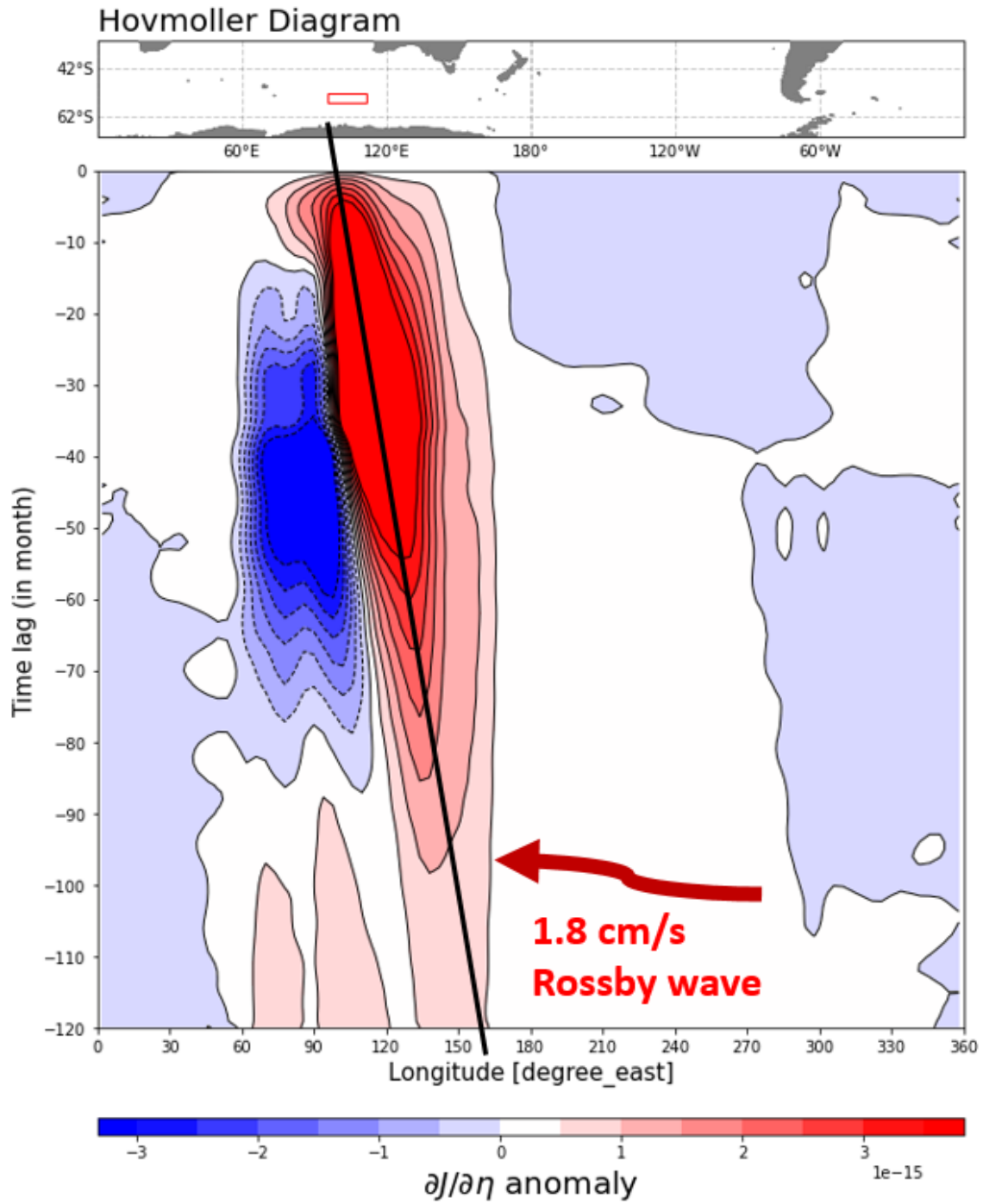


Figure 4.6: Hovmöller diagram for the the sensitivity to SSH anomaly ($\partial J / \partial \eta$). The unit is canceled due to the same unit of the objective function. The top panel shows the extent to which the Hovmöller diagram is evaluated and the box region in red.

4.2.2 Sensitivity to the buoyancy components

Buoyancy forcing is the sum of the net surface heat and freshwater fluxes. By convention, a positive surface flux represents flux upward from the ocean to the atmosphere. In other words, a positive heat flux means ocean heat loss (ocean cooling); A positive freshwater flux means ocean freshwater loss (ocean salinification), and vice versa. Figure 4.7 shows the pattern and pathway of sensitivity to the net surface heat ($\partial J/\partial Q_{net}$, subplots a-d) and freshwater ($\partial J/\partial E_{mpr}$, subplots e-h) fluxes. The sensitivities pattern for the two components is very similar, and their interpretation reflects similar effects.

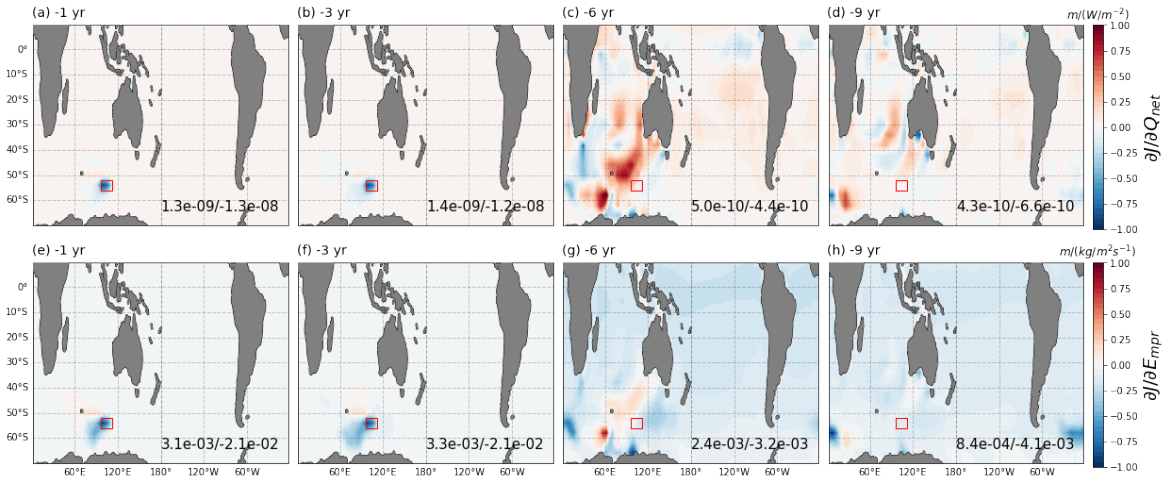


Figure 4.7: Evolution of the adjoint sensitivities (from left to right, backward in time) to surface net heat flux ($\partial J/\partial Q_{net}$, subplots a-d) and freshwater flux ($\partial J/\partial E_{mpr}$, subplots e-h). The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative sensitivity in each snapshot. The red box indicates the region where the objective function is defined.

At -1 and -3 years, the heat and freshwater flux sensitivities are confined to the box, with a slight shift to the west. The interpretation relates to the local steric effects: the strong negative sensitivities near the box indicate surface warming and freshening, respectively. The density of the water column decreases due to thermal expansion and direct freshening. Adding water regionally due to net precipitation and redistributing water masses due to density changes promote enhanced local SSH variability near the box region.

At -6 and -9 years, the negative sensitivities to the heat flux move westward to the south of Africa and the Southern Ocean Atlantic Sector. At the same time, the positive sensitivities to the heat flux are apparent in the Indian Ocean and coastal regions, with a comparable magnitude to negative sensitivities, suggesting the adjustment of surface warming and cooling. In addition, the alternation of positive and negative sensitivities in both hemispheres reveals a strong interannual heat flux variability (not shown, see animations in the supplementary). Nevertheless, a Hovmöller diagram (not

shown) demonstrates the strong negative, localized sensitivities in the Southern Ocean dominate and leave a net negative sensitivity propagating westward.

The sensitivities to freshwater flux are primarily negative almost throughout the whole domain, with dominant signals propagating in the ACC sector. The predominance of negative sensitivities and its pathway imply that ACC plays a significant role in transmitting surface salinity anomalies.

In short, one can notice that the propagation of the sensitivities to buoyancy forcing differs from momentum forcing in two ways. First, buoyancy has considerable centralized contributions with dominant negative sensitivity in the ACC sector at all other time lags, despite the effects of heat flux seasonality. Moreover, the leading negative sensitivities are mainly propagating in the opposite direction in the ACC region, contrary to the eastward propagation of momentum sensitivity.

4.3 Response to forcing

Note that the units of sensitivity fields to surface forcing may vary, preventing direct comparisons between atmospheric fluxes. In addition, while the adjoint sensitivity discloses when and where the target quantity can be optimally provoked, the target quantity's variability may only be observed if the forcing variability projects onto these sensitive regions. Therefore, it is required to scale the sensitivity fields by multiplying them with the variability of the forcing. The scaled sensitivities evaluate the response to forcing variability and their relative contributions. The response to the forcing variation emphasizes the practical information under the specific forcing condition.

Recall that the adjoint sensitivity is associated with the actual perturbations in the objective function via Taylor expansion in the vicinity of a background state:

$$J(x_i) = J(x_{i0}) + \left. \frac{\partial J}{\partial x_i} \right|_{x_0} \cdot (x_i - x_{i0}) + O(\|x_i - x_{i0}\|^2) \quad (4.1)$$

$$\Delta J = J(x_i) - J(x_{i0}) = \left. \frac{\partial J}{\partial x_i} \right|_{x_0} \cdot (x_i - x_{i0}) \quad (4.2)$$

The response of the objective function ΔJ can be approximated by multiplying the gradient (sensitivity) with the forcing anomalies. Such anomaly $(x_i - x_{i0})$ represents the expected uncertainty or variability in the forcing. The approximation holds valid so long as the anomalies remain small compared with the background state.

Different choices have been used to scale the sensitivities, such as forcing anomaly or standard deviation. One common choice used in previous studies (Heimbach et al., 2011; Verdy et al., 2014) is the two-dimensional, time-independent standard deviation fields of atmospheric forcing.

In the present study, the forward model is driven by monthly climatology because we do not consider simulating a specific event. The SSH in forward solution does not show internal variability apart from a damping oscillation resulting from the initial adjustment. We thus consider large-scale SSH variability contained in the monthly-varying forcings under global warming scenario.

The adjoint sensitivities are linearized about an equilibrium reference state. Thus, we expect the sensitivity pattern remains essentially unchanged under a strong warming scenario. This is verified by a forward model driven by alternative forcings representing the global warming scenario, and the sensitivities pattern virtually does not change.

To discover the underlying mechanism of low-frequency sea level variability under global warming, we use the standard deviation field of the last fifty years (2050-2100) of the RCP 8.5 surface forcings obtained from MPI-ESM. We applied a low-pass filter (period > 36 months) to capture the low-frequency variability in the forcings. The standard deviation field is shown in figure 4.8. Note, we only choose one member of the ensemble.

The scaled sensitivity or response to surface forcing can be expressed as:

$$dJ_x(\mathbf{r}, t) = \left[\frac{\partial J}{\partial x}(\mathbf{r}, t) \right] \sigma_x(\mathbf{r}) \quad (4.3)$$

where subscript x represents different forcing, $\sigma_x(\mathbf{r})$ represents the two-dimensional, time-independent standard deviation of the forcing component x after filtering out high-frequency variability. The scaled sensitivity $dJ_x(\mathbf{r}, t)$ can be obtained by multiplying with three-dimensional, time-varying sensitivity fields $\left[\frac{\partial J}{\partial x}(\mathbf{r}, t) \right]$ to each forcing variability $\sigma_x(\mathbf{r})$, making them comparable quantitatively. From another perspective, it represents the normalized sensitivities in response to the specific forcing condition.

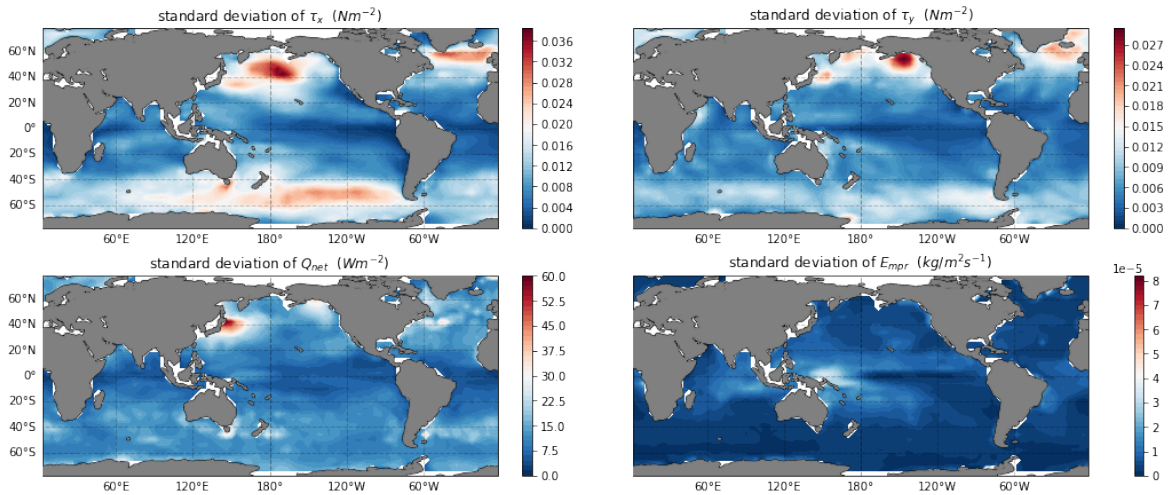


Figure 4.8: The standard deviation of surface forcings under RCP 8.5 scenario after applying 36 months low-pass filter. Strong magnitude in each subplot indicates the region with relatively high uncertainty for each forcing.

4.3.1 Scaled sensitivity

The scaled sensitivity embodies the combined effects of sensitivity and forcing variation. This may be enhanced if the sensitive region coincidence with the high variability region. Otherwise, the sensitivity pattern can be diminished if the sensitive region overlaps the low or zero variability region.

The distribution of scaled sensitivity to wind stress components, in general, resembles its original sensitivity map. However, the distribution of scaled freshwater sensitivity has been changed due to the relatively high variability in equatorial regions.

Figure 4.9 shows the scaled sensitivity to the surface wind stress. The previous analysis suggests the wind-sensitive region is primarily between Australia and Antarctica, and the zonal and meridional wind stress variability span a long band of ACC section. Thus, the scaled sensitivity to the wind stress may be enhanced compared with other regions. In contrast, the sensitivities to the zonal wind stress in the equatorial and Pacific Ocean seems weakened due to small low-frequency variability in these regions, indicating that the nonlocal planetary waves in these regions may be less significant under the global warming scenario.

Most importantly, the relative contribution of each forcing can be compared quantitatively. The sensitivity to the zonal wind is approximately one order of magnitude larger than that of meridional wind, highlighting the leading role of the westerlies jet over the Southern Ocean.

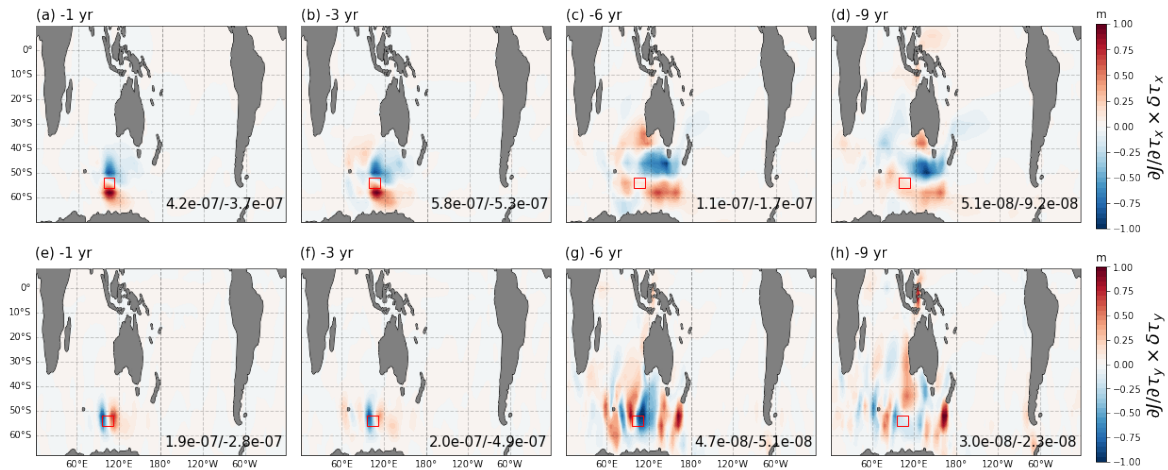


Figure 4.9: Evolution of the scaled sensitivity (from left to right, backward in time) with respect to zonal wind stress, scaled by 36-months lowpass filtered standard deviation of zonal wind stress ($\partial J / \partial \tau_x$, subplots a-d) and meridional wind stress ($\partial J / \partial \tau_y$, subplots e-h). The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative sensitivity in each snapshot. The red box indicates the region where the objective function is defined.

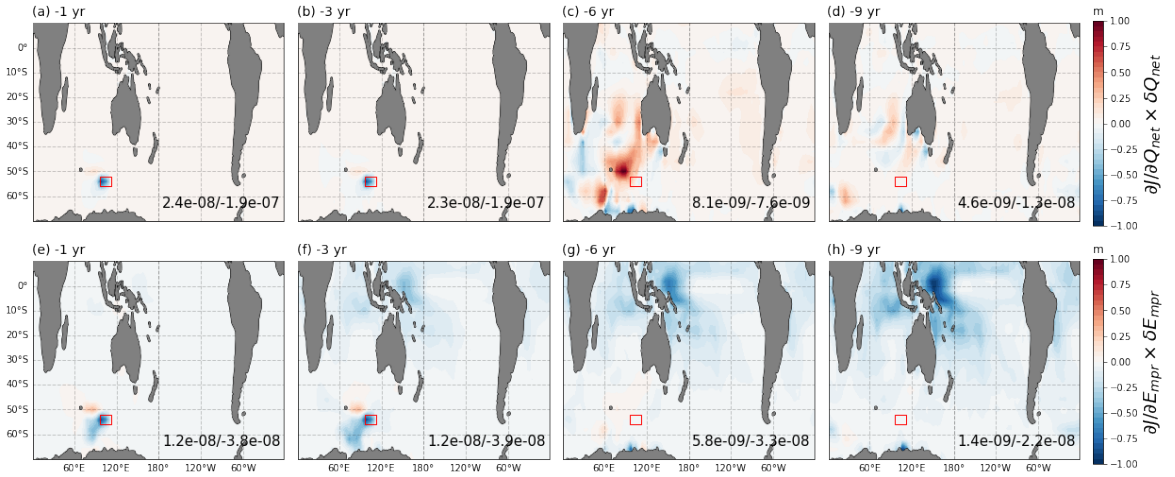


Figure 4.10: Evolution of the scaled sensitivity (from left to right, backward in time) with respect to buoyancy components, scaled by 36-months lowpass filtered standard deviation of surface heat flux ($\partial J/\partial Q_{net}$, subplots a-d) and freshwater flux ($\partial J/\partial E_{mpr}$, subplots e-h). The sensitivities are normalized by their maximum magnitude to highlight the most dominant signal, as labeled in each subplot showing the maximum positive/negative sensitivity in each snapshot. The red box indicates the region where the objective function is defined.

Figure 4.10 shows the scaled sensitivity to the buoyancy components. Scaled heat flux sensitivity distribution is almost unchanged compared with the adjoint sensitivities because the low-frequency variability of heat flux forcing is not varying in the Southern Hemisphere.

In contrast, the distribution of scaled freshwater sensitivity has changed significantly, reflecting the freshwater variability in the equatorial and coastal shelf seas. Its magnitude is lesser than heat flux sensitivity but spreads a much broader region, suggesting that the contribution of freshwater may have a comparable influence with heat flux.

Overall, the net scaled heat and freshwater flux sensitivities dominate the Southern Ocean AAC sector, meaning the primary response to buoyancy components is also in the ACC sector, besides the freshwater contribution in the Western Pacific. Meanwhile, the positive sensitivities reflecting seasonality may have only a limited influence.

4.3.2 Momentum versus buoyancy

Following Verdy et al. (2014) and Jones et al. (2018), one can define a measure of relative response in order to compare each attribution. We adopt the summation in time of the absolute values for each component, quantifying the magnitude of spatially varying sensitivity (dJ_{var} , Eq. 4.4). This measure avoids the cancellation of positive and negative sensitivities. The total response of the objective function to each of the four surface forcing fields can be expressed as:

$$dJ_{var} = \langle |dJ_{\tau_x}| \rangle + \langle |dJ_{\tau_y}| \rangle + \langle |dJ_{Q_{net}}| \rangle + \langle |dJ_{E_{mpr}}| \rangle \quad (4.4)$$

where vertical bars represent absolute value, and angular brackets indicate the average over a chosen domain. Since the dominant sensitivities are confined to the Southern Ocean Indian Sector, it is not necessary to partition the domain into various sections, thus we consider the global domain.

The total scaled sensitivity is summed over space and time, and the relative contribution of each forcing to the dJ_{var} are summarized in the table 4.1. It shows that the objective function is most sensitive to the zonal surface wind stress, which contributes 48% of the total value, and meridional wind stress contributes 19 %. On the other hand, buoyancy components also play a role, the heat flux accounts for 13%, and the effect of freshwater flux is about 20%.

Table 4.1: **Variance explained by each component**

Forcing	Summed dJ_{var}
Zonal wind stress	48%
Meridianl wind stress	19%
Net heat flux	13%
Net freshwater flux	20%

4.3.3 Local versus remote

(a) spatial distribution

Figure 4.9 suggests that the dominant influences of the winds are primarily located between Australia and Antarctica over the whole period. However, nonlocal effects accumulate more slowly and are conceivable at remote time lags. The nonlocal influence of the zonal wind can be noticed in the tropical and mid-latitude oceans. In addition, the impact of the meridional wind is especially seen along the continental coasts and the Indonesian Passage.

Figure 4.10 indicates that the influence of the two buoyancy components is widely distributed in the open oceans and ACC sector. The net heat flux has a strong seasonality suggested by the alternation of adversary signs in both hemispheres. However, its impact is less significant than the local centralized negative sensitivities in the ACC section. In contrast, freshwater sensitivity spreads more broadly, with dominant sensitivities in the Western Pacific and coastal shelf seas. It is conceivable to have an even more significant effect than net heat flux when the scaled sensitivities are summed over the global domain.

Figure 4.11 depicts the spatial locations of scaled sensitivities when summed over ten years. Wind stress components and buoyancy components are superimposed, respectively. It reveals that zonal wind stress is the leading contributor to momentum components. In contrast, the effects of heat and freshwater flux can be superimposed in the

ACC sector, as both show dominant negative sensitivity there. While in the non-local open oceans, effects of positive heat flux sensitivity cancel with negative freshwater sensitivity, as they show adversary signs.

In short, the response to momentum components is concentrated between Australia and Antarctica, downstream of the ACC sector, close to the east side of the box. In contrast, the response to buoyancy components is much localized upstream of ACC, on the west side of the box.

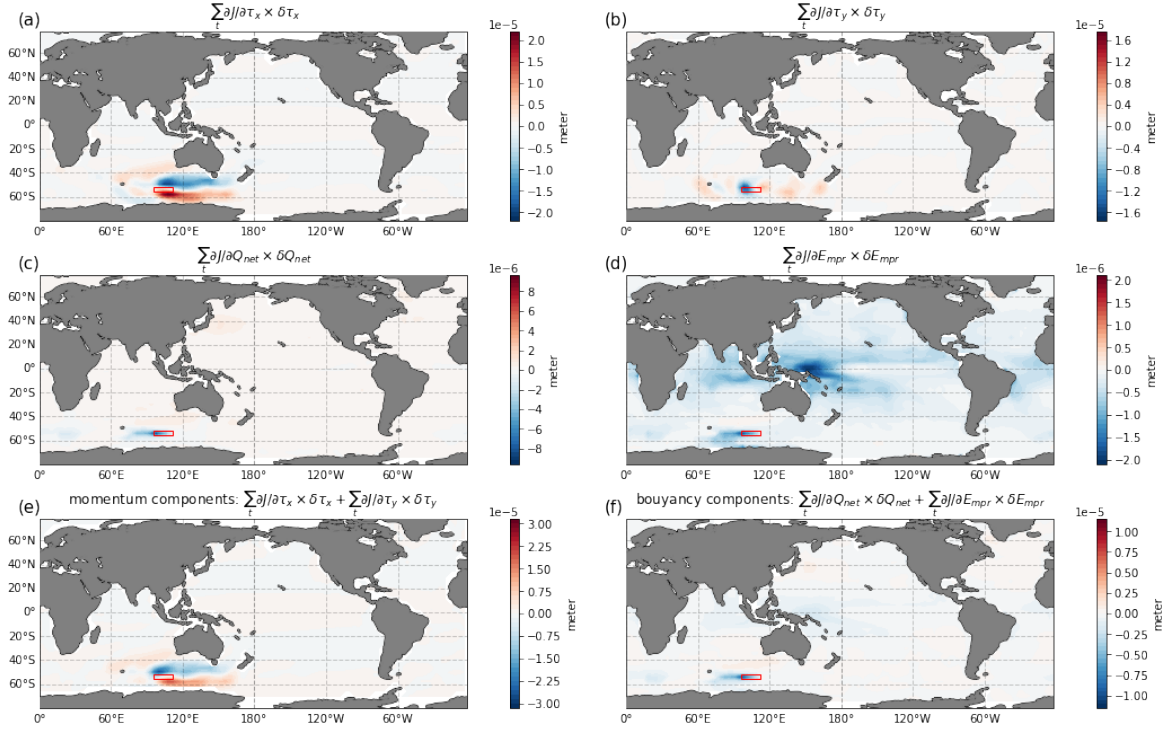


Figure 4.11: 10-year temporal integration of the scaled sensitivity. (a) with respect to the zonal wind stress, (b) meridional wind stress, (c) net heat flux, and (d) net freshwater flux. Subplot (e) represents a superposition of surface zonal and meridional wind stress sensitivities and (f) a superposition of surface net heat and freshwater flux sensitivities. All subplots have a unit of meter.

(b) temporal evolution

The sensitivity can also be examined in time dimension by summing over space. As shown in Figure 4.12, the spatially summed scaled sensitivity time series demonstrates the temporal evolution of response magnitude to each forcing component.

In a backward order (view the figure from right to left), as expected, the absolute sensitivity accumulates during the objective function's evaluation period and reaches the maximum at around -3 years, then gradually decays backward in time.

The zonal wind is the leading forcing at all times, reflecting that the objective quantity is most sensitive to the zonal wind stress. The evolution of scaled absolute sensitivity to

the meridional wind is similar to the zonal wind but with a smaller magnitude. Their evolution is mainly linear, while the buoyancy components demonstrate the annual oscillation, especially the heat flux, reflecting the interannual changes of ocean warming and cooling in austral winter and summer.

From 0 until -5 years, the wind stress components exhibit a greater magnitude than the buoyancy components, suggesting that wind-driven processes are mainly responsible for low-frequency, large-scale sea level variability.

From -3 to -10 years, the magnitude of heat and freshwater flux sensitivities remain relatively stable and less significant compared with wind stress components, suggesting buoyancy-induced effects must ultimately accumulate from the nonlocal regions to the objective region, which takes several years. Furthermore, heat and freshwater flux curves show a slight opposite tilt, reflecting their relative importance to the objective quantity at different time lags.

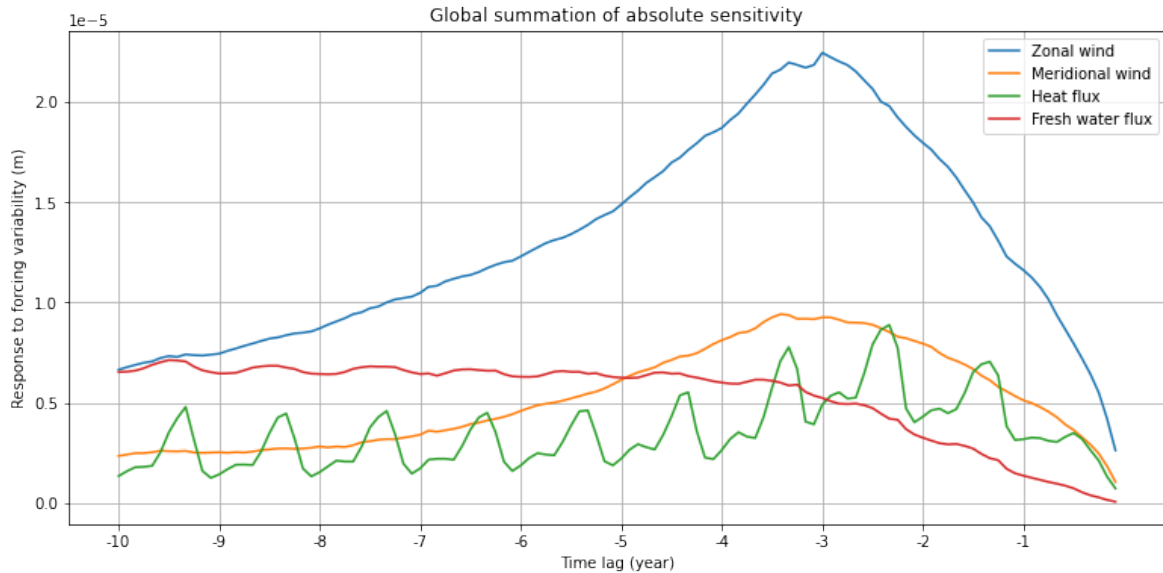


Figure 4.12: Temporal evolution of absolute scaled sensitivity for the zonal wind stress, meridional wind stress, surface heat flux, and freshwater flux.

4.4 Perturbation experiments

As mentioned in Chapter 3, the adjoint model is a powerful tool for estimating the sensitivity of the objective quantity to input forcing through backward propagation. On the other hand, one can further verify the validity of linear assumption and emphasize the underlying physical processes encoded in the model through the forward perturbation experiment based on the obtained adjoint sensitivity.

4.4.1 Wind stress perturbation

From previous sensitivity analysis, wind stress contributes most of the objective function response. Now, we describe the procedures of wind stress perturbation experiments and discuss how the perturbation experiment can shed light on the underlying mechanism.

Firstly, one needs to know the perturbation pattern. Following Wang et al. (2022), we adopt the sensitivity-like perturbation, which means the pattern of the wind stress perturbation resembles the adjoint sensitivity to the wind stress. To constrain the upper magnitude of the perturbation, we first normalize the adjoint sensitivity to the zonal wind stress by its maximum absolute value and then set it to be proportional to the default wind stress by a factor of δ . Likewise, the meridional perturbation is proportional to zonal wind perturbation. Intuitively, the wind perturbation fields can be expressed via mathematical equations:

$$\Delta U = \frac{\partial J / \partial \tau_x}{\max(|\partial J / \partial \tau_x|)} \cdot \delta \cdot \max(|U_0|) \quad (4.5)$$

$$\Delta V = \frac{\partial J / \partial \tau_y}{\partial J / \partial \tau_x} \cdot \Delta U \quad (4.6)$$

Where U_0 represents the original zonal surface wind stress. The choice of δ usually reflects the magnitude of forcing uncertainty, here δ is set to be 20% for all perturbation experiments.

Secondly, considering the temporal evolution of sensitivity, we designed an experiment by imposing sensitivity-like wind perturbation in the first three years (in a forward sense), meaning the sensitivities are from -10 to -8 years (in a backward sense). Precisely, the time-dependent wind stress perturbation is superimposed on the original wind stress field and only lasts over the first three years. Figure 4.13 shows the mean sensitivity-like wind stress perturbation imposed over three years, in which its distribution resembles sensitivity to the wind stress.

Finally, we integrate the model forward for ten years under the same configuration. The only difference is the wind stress perturbation. Then we compare the model trajectory regarding SSH and zonal surface velocity by subtracting the trajectory of the reference run. Figure 4.14 presents the response of SSH and zonal velocity in the first, third, sixth, and ninth years (corresponding to -9, -7, -4, and -1 years in the backward sense).

From -10 to -8 years, the equatorial positive wind stress perturbation accelerates zonal surface velocity in the first three years. As a result, it produces positive SSH anomalies at the equator, characterizing equatorial Kelvin waves propagating eastward and generating coastally trapped waves poleward at the continent boundaries. At the same time, an anticyclonic wind stress shear in the Southern Ocean Indian Sector generates

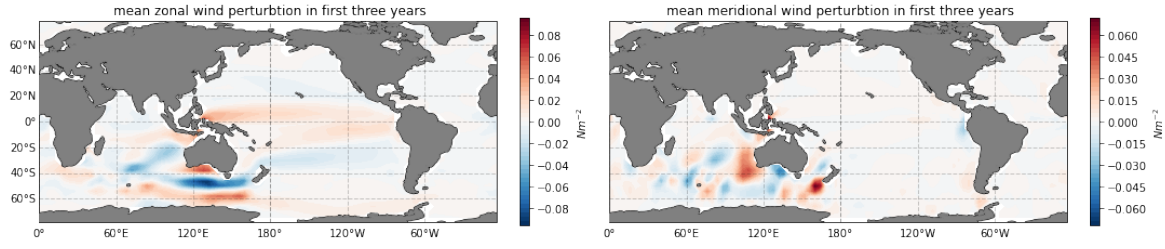


Figure 4.13: Mean sensitivity-like wind perturbation imposed in the first three years, with a magnitude of 20% of the surface wind stress. Note the perturbation is time-dependent, following the same temporal evolution of sensitivities. A positive zonal wind stress perturbation means an increase in eastward zonal wind stress.

local positive SSH anomalies immediately incited by local wind stress curl, with the maximum at the east of the box.

At -4 years, the equatorial Kelvin wave has reflected as Rossby waves in both hemispheres, which exhibit latitude-dependent phase speed variation due to the β effect. The zonal surface velocity in the Pacific basin is reduced due to the westward-propagating Rossby waves, which gradually reach Australia's east coast, passing through the Tasman sea and enhancing positive SSH anomaly south of Australia. In the Southern Ocean Indian Sector, the positive SSH anomalies have propagated to the west in the form of the baroclinic Rossby wave.

At -1 year, the perturbed SSH ultimately arrived around the target box, with joint contributions from wind perturbation over the Southern Ocean and the equator. The response to the wind South of Australia is almost immediate. However, the response to equatorial wind perturbation takes around 7-9 years to accumulate and reach the target region. The timing and pathway of signal propagation are consistent with the interpretation of adjoint sensitivity.

While the magnitude of wind perturbation is most prominent in the Southern Ocean, the corresponding changes in zonal surface velocity are less significant than at the equator. It reflects that the slow westward propagation of Rossby wave is Doppler-shifted through advection with the ACC mean flow.

To further confirm the relative contribution between wind in the Southern Ocean and equator, we repeat separate perturbation experiments, in which we apply wind stress perturbation confined within and out of the Southern Ocean, representing the effects of local wind over the Southern Ocean and nonlocal wind from the equator, respectively.

Again, we compare the difference in SSH anomaly with the reference run (see Fig 4.15). For the nonlocal wind perturbation, the response on our target region takes 7-9 years to arrive. The positive SSH difference only shows up over the last three years. For the local wind perturbation, the response on the target region is immediate, generating positive SSH anomalies over a large area in the Southern Ocean Indian Sector.

Nevertheless, both perturbation experiments generate positive SSH anomalies during

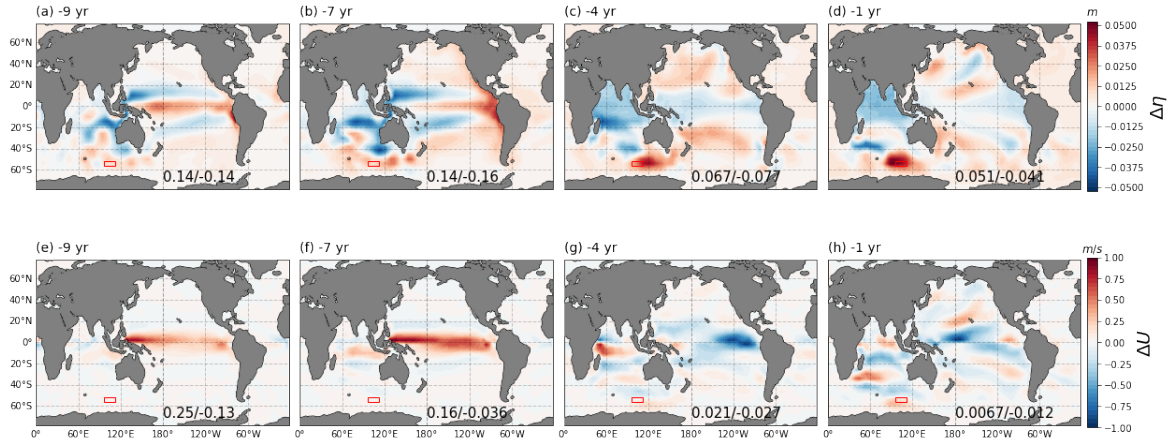


Figure 4.14: Response (perturbation run minus the reference run) of SSH anomaly ($\Delta\eta$, subplots a-d) and zonal surface velocity (ΔU , subplots e-h), at -9, -7, -4 and -1 year.

the objective period. The perturbation experiments prove the meaning of the adjoint sensitivity: impose a sensitivity-like perturbation in control variables at previous states, and the response of the objective function will increase accordingly.

Furthermore, we see different patterns for perturbations at different locations, suggesting the mechanisms evoked by nonlocal and local wind. For example, the local wind perturbation generates Ekman pumping or suction incited by wind stress curls. In contrast, nonlocal wind perturbation generates equatorial Kelvin waves, evolving into coastally trapped waves and reflected Rossby waves.

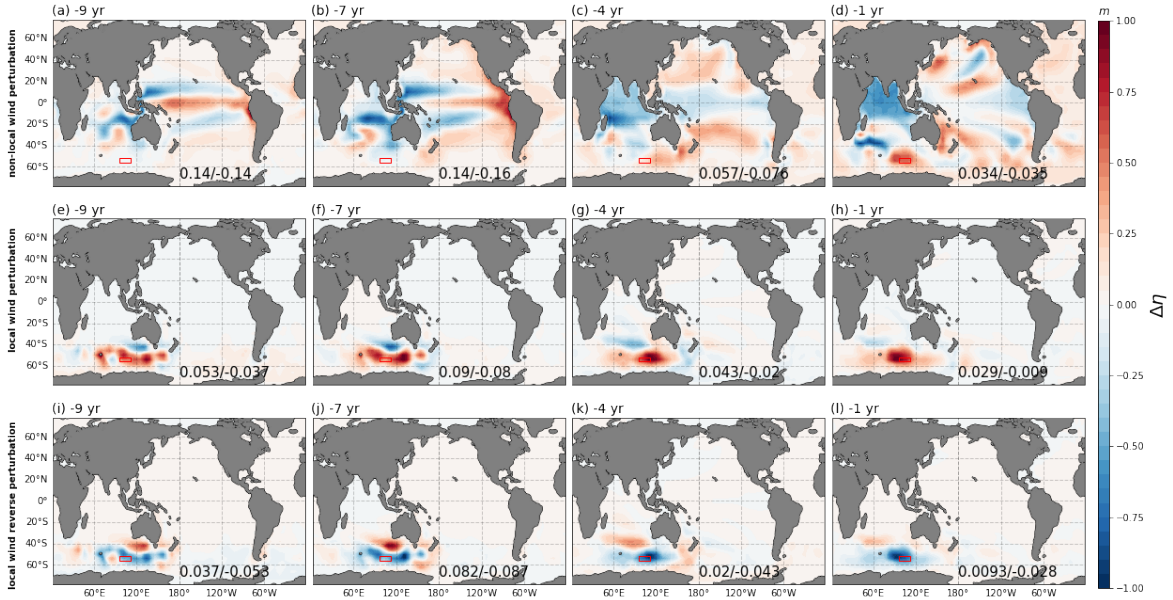


Figure 4.15: Response of SSH anomaly (SSH anomaly in perturbation run minus the reference run). The first row shows the response when imposing non-local wind perturbation (subplots a-d), the second row imposing local wind perturbation (subplots e-h), and the third row imposing reversed local wind stress perturbation (subplots i-l).

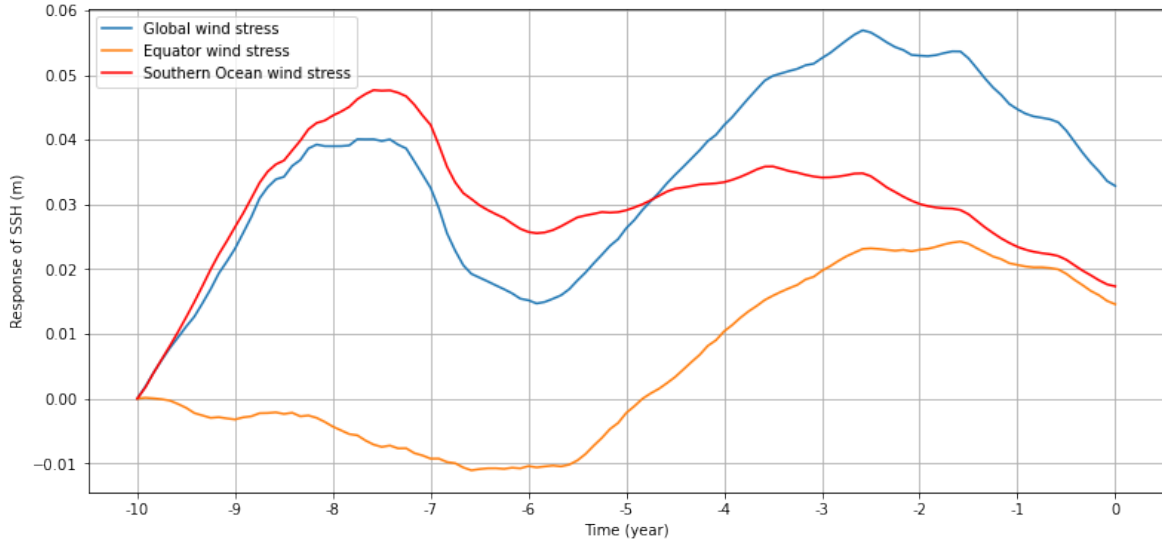


Figure 4.16: The spatial mean SSH anomalies in the box response to the global, Southern Ocean, and equator wind stress perturbations in the first three years. The global wind perturbation is a combination of Southern Ocean and equator wind stress perturbation.

Figure 4.16 shows the time series of the SSH anomalies response in the box region for surface wind stress perturbation indicated by the legend. The response to the Southern Ocean wind stress perturbation promotes positive SSH anomalies, which are immediate and robust in the first three years (-10 to -7 yr). It declines when the perturbation stops, rises to a maximum at about -2.5 years, and then gradually decays. In contrast, the response to the equatorial wind stress perturbation is negative over the first five years (-10 to -5 yr). It then gradually increases at around 5th year, meaning the positive SSH anomalies induced by the equatorial wind stress take at least five years to be effective.

Additionally, the SSH response to the global wind perturbation can be split into different regions. We obtain a response that almost matches the global wind perturbation by superimposing the responses to the Southern Ocean and equator wind stress perturbation, which implies the SSH response to sensitivity-like perturbation is almost linear.

4.4.2 Heat flux perturbation

The surface heat flux is a minor factor contributing to the objective function response. Here, we adopt a similar perturbation experiment and briefly present the SSH response to distinguish it from wind stress perturbation experiments. Like previous wind stress perturbation experiments, we use the same formula to prescribe sensitivity-like heat flux perturbation in the first three years.

$$\Delta Q = \frac{\partial J / \partial Q_{net}}{\max(|\partial J / \partial Q_{net}|)} \cdot \delta \cdot \max(|Q_{net}|) \quad (4.7)$$

Where Q_{net} represents the original surface heat flux, and δ is set to be 20%.

Figure 4.17 presents the response of SSH anomaly and SST. As indicated by the sensitivity, sensitivity-like heat flux perturbation promotes positive SSH and SST anomalies, which propagate along ACC in the Southern Ocean. The positive SSH anomaly has occupied the box in the last three years. However, the response magnitude is less significant than the wind stress perturbation experiment, as the objective function is less sensitive to heat flux than zonal wind stress.

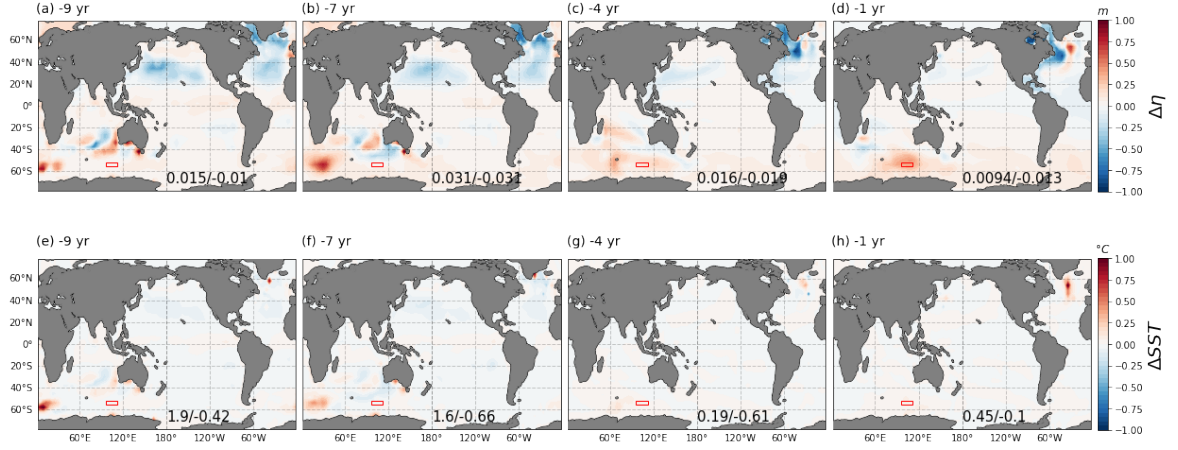


Figure 4.17: Response (perturbation run minus the control run) of SSH anomaly ($\Delta\eta$, subplots a-d) and SST (ΔSST , subplots e-h), at -9, -7, -4, and -1 year.

Chapter 5

Discussion

This chapter first discusses physical mechanisms driven by surface wind stress and buoyancy flux. In addition, we deliver our perspectives in light of the global warming scenario. Finally, we reflect on the validity of this research and discuss its limitation.

5.1 Wind-driven mechanism

The changes in the surface wind stress fields determine regional dynamic sea-level variation through Ekman pumping or suction, coastal upwelling, or wave propagation.

The mechanism of wind-driven sea level variation can be explained as follows. The strengthening of anticyclonic wind stress in the east of the box (a positive wind stress curl) promotes the Ekman convergence of surface water, thus, Ekman pumping in the upper layers. The wind stress curl drives the westward-propagating baroclinic Rossby waves, which may be enhanced by another planetary Rossby wave from the subtropical ocean and Australian coastal waves through the Tasman Sea. The enhanced Rossby wave propagates westward, even though it is advected by the eastward ACC. The rising sea levels move westward, amplifying the low-frequency SSH fluctuation in the Indian Sector of the Southern Ocean.

The adjoint sensitivity suggests that local wind stress is the primary contributor. Furthermore, the SSH responds immediately to wind stress perturbations over the Southern Ocean. In contrast, nonlocal winds in the tropics contribute a small fraction, but the response takes approximately 7–9 years to arrive. The tropics' nonlocal winds promote an eastward-propagating Kelvin wave, which is trapped along the South American coast. It gradually reflected as a westward-propagating Rossby wave in the Southern Hemisphere, passing through the Tasman Sea and enhancing the SSH south of Australia. However, the nonlocal wind-driven mechanism may be diminished if wind stress variability in the Pacific Ocean is inconsequential.

5.2 Buoyancy-driven mechanism

The steric sea level variations are changes in ocean volume induced by density changes, which can be split into thermostreic and halosteric components. The surface heat and freshwater fluxes influence the density in the upper ocean through thermodynamics. The resulting changes in ocean density promote steric sea-level variations.

The role of thermostreic effects can be estimated from the temperature field, which is indicated by the sensitivity to the surface net heat flux. The sensitivity pathway reveals surface ocean heat content redistribution, which is modulated by interannual ocean warming and cooling. Over a decadal timescale, the leading negative sensitivities spread along the ACC, propagating temperature and salinity anomalies. Negative sensitivities mean surface ocean warming. Surface temperature increases lead to thermal expansion and an increase in water volume. Note that the positive sensitivities may prevail seasonally, suggesting ocean cooling and salinification can compensate for a part of thermal expansion-induced sea level change.

The role of halosteric effects can be estimated from the salinity field, which is often described in terms of freshwater content. The addition and subtraction of freshwater input directly change the water volume and salinity. The changes in density can further affect ocean circulation. Significant freshwater variability is primarily distributed in the west Pacific under the RCP 8.5 scenario. The scaled freshwater sensitivities indicate that, in addition to the Southern Ocean Indian Sector, the west Pacific is a sensitive region. The effects of freshwater content in the west Pacific may be propagated by wind-driven circulation, such as the planetary eastward Kelvin wave, or topography-steered flow, such as the Indonesian throughflow.

5.3 Validation and limitation

Though it is advantageous compared with the forward method, adjoint sensitivity is the first-order derivative or tangent linear approximation to the actual sensitivity. The true sensitivity must have some degree of nonlinearity. Nonetheless, the linearity assumption for sensitivity analysis has been validated using coarse resolution models, with an inaccuracy being within 30% on a 100-year timescale and within 5% on the year scale (Losch and Heimbach, 2007).

Forward perturbation experiments can examine the linearity validation with small but sufficient positive and negative perturbations (Jones et al., 2018; Zhan et al., 2018). Specifically, imposing a forcing perturbation, the response of the objective function J can be approximated via Taylor expansion:

$$\Delta J = J(x_i) - J(x_{i0}) + \left. \frac{\partial J}{\partial x_i} \right|_{x_0} \cdot (x_i - x_{i0}) + O(\|x_i - x_{i0}\|^2) \quad (5.1)$$

We denote the response as ΔJ_+ if the perturbation is sensitivity-like and ΔJ_- if the perturbation is reverse-sensitivity-like (reverse the sign the sensitivity). The first derivative or linear term can be calculated as,

$$\Delta_1 \approx \frac{J(x_+) - J(x_-)}{2\delta x} \delta x = \frac{\Delta J_+ - \Delta J_-}{2} \quad (5.2)$$

and the second quadratic or nonlinear term can be estimated as,

$$\Delta_2 \approx \frac{\Delta J_+ + \Delta J_-}{2} \quad (5.3)$$

Based on the Southern Ocean wind stress perturbation experiment introduced in the previous chapter, we additionally conduct another investigation by imposing reversed wind stress perturbation. It means the positive perturbation becomes negative and vice versa. The SSH anomalies response to prescribed perturbation is shown in Figure 4.15, which produces a reversed effect compared with sensitivity-like perturbation. We obtain the linear and nonlinear response of SSH anomaly in the box region using the formulas above, as shown in Figure 5.1. As expected, the large-scale SSH response to the imposed perturbation is almost linear over ten years of integration. Only a fraction of non-linearity exists in the 5-7 th year.

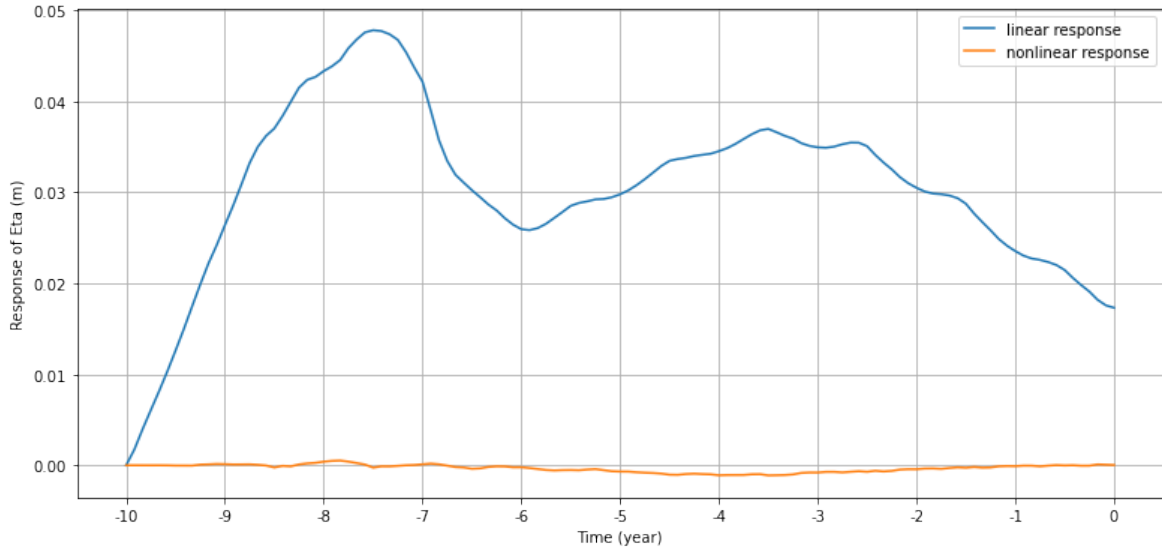


Figure 5.1: Linear and nonlinear response of SSH anomaly in the mode region to Southern Ocean wind stress perturbation.

The linear assumption holds valid as long as the linear response is dominating compared with the nonlinear response. It also reveals how long the linear assumption holds valid. However, cautions need to be made when nonlinearity is significant.

Below, we list a few limitations of the adjoint-based sensitivity analysis in this study:

1. Ocean-only model.

The influence of the atmosphere may need to be more conclusive. The driving forcings are integrated ocean surface fluxes but do not separate specific atmospheric variables, including radiation and air temperature. How the ocean responds to the changes in the atmosphere needs to be clarified.

One can argue that the influence of melting sea ice and glacier from Antarctica may be limited in the region of our interest in the open ocean. Even though the effects of sea ice are symbolized in the freshwater flux, significant uncertainties remain due to a limited understanding of sea ice dynamics. An ice-coupled model might be insightful to address the concerns.

2. Coarse resolution.

We configured a very coarse resolution in a balance between accuracy and efficiency. On the one hand, it seems reasonable to filter out small-scale, high-frequency signals, which can be regarded as noises for large-scale, low-frequency oceanic variability. On the other hand, the effects of nonlinear eddies on the sea level variability are worth further investigation because eddies are dominant in the Southern Ocean. (Morrison and Hogg, 2013) claims eddy compensation and saturation can offset the influence of surface wind stress change.

Losch et al. (2004) suggests the sensitivity patterns remain largely the same when the resolution is doubled from 4° to 2° , but some detailed features, especially in boundary regions, may change the sign. Jones et al. (2018) indicates some deep ocean properties in a 1° model (ECCOV4) is in broad agreement with the $1/6^\circ$ eddy-permitting Southern Ocean State Estimate (SOSE). Care must be taken when interpreting the sensitivity of a specific studying object. The sensitivity of oceanic properties to model resolution may be another valuable topic to investigate. In addition, the ocean dynamics may not be well represented in topography-influenced regions, such as the Indonesian Passage.

3. Adjoint limitation.

The linearity assumption in the adjoint technique is a double-edged sword. The adjoint sensitivity has absolute advantages over the conventional sensitivity in terms of efficiency but loses an intrinsic ocean property, nonlinearity. Therefore, the interpretation is only applicable when the linearized assumption is valid for the object of interest, which the nonlinear perturbation experiment can further verify. Moreover, some parameterization may be sacrificed in exchange for linearity. The current model configuration simplified some processes in the deep ocean, such as the unexploited eddy parameterization. Hence, a further study on how the model parameterization affects the adjoint sensitivity is constructive.

Chapter 6

Conclusions and outlook

The present study is a sensitivity analysis using MITgcm and its adjoint model. In addition to interpreting the derived linear sensitivity, we project the forcing variability of the RCP 8.5 scenario into sensitivity fields to assess their relative importance. Furthermore, we employ sensitivity-like perturbation experiments, highlighting the physical processes indicated by sensitivity. Here we summarize the significant results based on sensitivity analysis. In the end, we offer a list of perspectives on potential extensions of this work.

6.1 Sensitivity analysis

We use MITgcm and its adjoint model to investigate the linear sensitivity of low-frequency regional sea level variation in the Southern Ocean Indian Sector to surface forcings, including surface zonal wind stress, meridional wind stress, net heat flux, and freshwater flux. Below we draw a schematic to summarize the underlying mechanism indicated by the propagation of sensitivity fields (See Fig 6.1).

The nonlinear forward model simulates a realistic global ocean and establishes the background state. In contrast, the adjoint model perturbs SSH in the target box over the last three years and propagates the model physics backward in time, producing the gradient of the objective function for all control variables and model states at earlier times with one integration. The first-order gradient, known as the adjoint sensitivity, reveals the physical mechanism and separates the different effects of each forcing component, as indicated by its spatial distribution and temporal evolution.

The sensitivity to surface wind stress suggests that a wind-forced Baroclinic Rossby wave is the primary causal mechanism for the low-frequency sea level in the target region. The wind-sensitive region is on the east side of the box, between Australia and Antarctica, where the intensified anticyclonic surface wind stress drives Ekman pumping and a westward-propagating Rossby wave. Over a decadal timescale, the

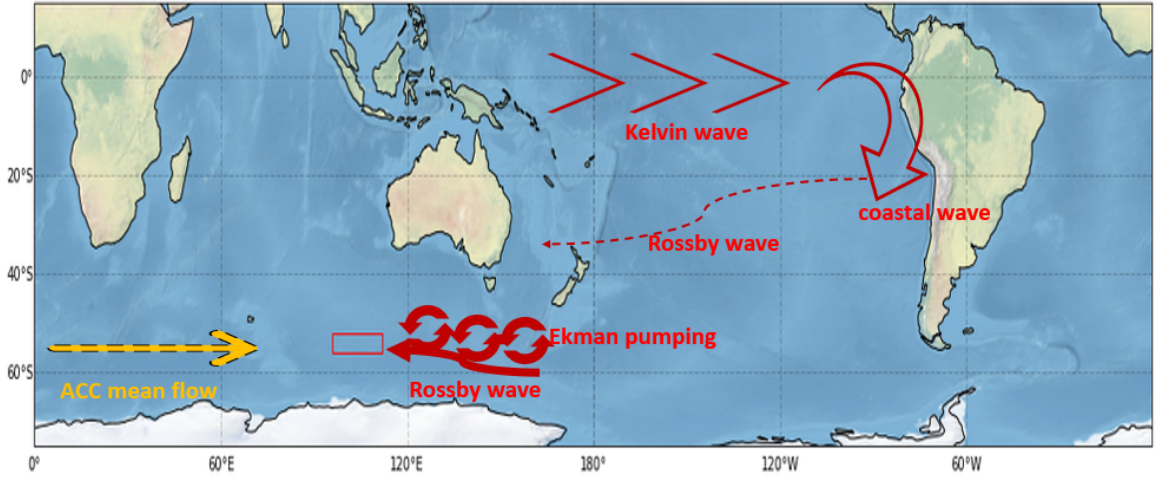


Figure 6.1: Schematic of underlying oceanic processes that affect the objective function, indicated by the global wind stress and heat flux perturbation experiments. The red box indicates the region where the objective function is defined.

Rossby wave shifts the Ekman pumping westward. The SSH response is immediate, but it is slowed by advection with ACC mean flow, which is known as the Doppler-shift effect. Moreover, the nonlocal wind stress from the equator may also contribute a small part if forcing variability is projected into these regions. The zonal wind stress drives the eastward Kelvin wave and is reflected as the Rossby wave, enhancing SSH variability south of Australia.

The sensitivity to surface buoyancy components propagates along the ACC mean flow, and the steric effects can explain their contribution to sea level. A further investigation on the sensitivity to potential temperature may reveal more information, as previous studies have split the sensitivities into kinematic and dynamic parts (Marotzke et al., 1999). The effects of buoyancy components accumulate for several years before arriving at the target region, and their pathway reflects the most important transport in the Southern Ocean.

We project the forcing variability into its corresponding sensitivity field to estimate the relative contribution of each forcing and probe the mechanism of sea level variability under the global warming scenario. We find that the sea level in the target region is primarily wind-driven, with the zonal wind stress accounting for approximately 48% of the total contribution and the meridional wind stress taking up 19%. On the other hand, the buoyancy components account for about 33%. The halosteric sea level change due to the influx of freshwater is more critical than the thermosteric sea level change caused by surface heat flux.

We further performed sensitivity-like perturbation experiments to verify the interpretation of the adjoint sensitivity and examine the linearity assumption. In addition, the perturbation experiments highlight the causal mechanism induced by the perturbation. For instance, the sensitivity-like wind stress perturbation reveals the eastward Kelvin

wave at the equator, the reflected westward Rossby wave in the Pacific basin, and the barotropic Rossby wave south of Australia. In contrast, the sensitivity-like heat flux perturbation reflects the ACC mean flow. Such sensitivity-like perturbations imposed on the background climatological forcings can optimally generate SSH anomalies in our target region.

6.2 Outlook

Following the conclusion from the adjoint sensitivity, we provide a few constructive outlooks which can further extend this work.

1. Ensemble.

The external forcings under the global warming scenario inherit certain uncertainties. We only used one member output from MPI-ESM, which cannot necessarily represent the majority statistically. Therefore, it is recommended to statistics of the ensemble.

2. Hindcasting/forecasting.

In the current study, we highlight the effects of surface fluxes on large-scale regional sea level change but do not construct sea level variability. Previous studies suggest the sea level variability can be reconstructed through the convolution with forcing anomalies (Fukumori et al., 2007, 2015, 2021).

3. Data-constrained, high resolution, coupled model.

The present work is a test bed to investigate large-scale upper ocean circulation using the adjoint model. We adequately estimated the causal mechanism of sea level variation based on sensitivity analysis. However, more accurate causal attribution can be reached with the advancement of observation and model development. More importantly, the contributing variables may not only be limited to the surface fluxes but more specific variables in the Atmosphere, Ocean and Ice. The ECCO consortium currently leads to achieving this ambiguous goal. More applications using the adjoint are ready to come in the coming years.

Supplementary

The model setup used in this work is available from https://github.com/MITgcm/MITgcm/tree/master/verification/global_ocean.90x40x15, as an example of the open source MIT general circulation model (MITgcm, <http://mitgcm.org/>)

The animations of sensitivity fields and some post-processing scripts are available at <https://github.com/oceanography-rookie/Adjoint-sensitivity>.

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Acronyms

ACC	Antarctic Circumpolar Current
AD	Algorithmic/Automatic Differentiation
AVISO	Archiving, Validation and Interpretation of Satellite Oceanographic data
CMIP	Coupled Model Intercomparison Project
DKRZ	Das Deutsche Klimarechenzentrum
ECCO	the Estimating the Circulation and Climate of the Ocean
EOF	Empirical Orthogonal Function
ETOPO5	Earth Topography Five-Minute Grid
GMSL	Global Mean Sea Level
HPC	High Performance Computer
IPCC AR6	the sixth Assessment Report of the Intergovernmental Panel on Climate Change
MITgcm	Massachusetts Institute of Technology general circulation model
MOC	Meridional Overturning Circulation
MPI-ESM	Max Planck Institute Earth System Model
MPI-GE	Max Planck Institute Grand Ensemble
OGCM	Ocean General Circulation Model
RCP 8.5	Representative Concentration Pathway 8.5
SAM	Southern Annular Mode
SSH	Sea Surface Height
SST	Sea Surface Temperature
TAF	the Transformation of Algorithms in the Fortran compiler
TAMC	Tangent linear and Adjoint Model Compiler
TLM	Tangent Linear Model

Acknowledgements

Working on this fascinating topic has been an incredible journey. Dr. Armin Köhl, my supervisor, inspired me to conduct this research. His patience and guidance were the most valuable assets to this thesis. I am grateful to my supervisor Prof. Dr. Detlef Stammer for offering an opportunity to work in his group and providing supports. They generously granted me access to the commercial software TAF, a necessary tool for this study. In addition, I appreciate tremendous help from Dr. Nuno Serra. His clear explanation and kind comments are indispensable parts of my thesis. His words relieved my anxiety and let me concentrate on thesis writing. He is not technically my supervisor, but he deserves my wholehearted appreciation. They are genuine role models as scientists from whom I learned to work and think independently. Their hardworking and dedication to climate science motivate me all the way.

I thank Dr. Lars Czeschel for an enjoyable discussion. He kindly shared his experience of adjoint modeling. I thank Dr. Sergiy Vasylevych for his friendly explaining my doubts. He is constantly being approachable to students. I thank Dr. Peng Zhan for kindly offering suggestions and encouragement at some points in my studies. I thank Dr. Yi Jin for offering constructive comments on my thesis.

I want to express my gratitude to all colleagues I have been bothered with, Chavaly, Chathurika, and everyone I have talked to within the group. I thank Dharmesh, Nikhil, Kanyan, Xiaoxue, and Philip for accompanying me. I thank our secretary Maite for helping with some nuances.

I am fortunate to have a native speaker, Philip, in my group. He spent some time and effort reading my thesis and offering great suggestions on linguistics. His help significantly improved the readability of my thesis.

I cherish a wild memory with my favorite friends during my abroad semester in the Netherlands, Ryan, Gary, Frankie, Tobias, Enej, and Patricia. Their companies made me feel at home in a foreign land. My time with them was the happiest of the past two years. They are always encouraging me along the roads ahead, even though we are far apart from each other.

I want to thank my friend Joshua. He was a Ph.D. student in oceanography, and we had just randomly met at a festival. He stayed in my office for some time, and his humor made these days delightful.

I want to thank my best friend, Niels Behrmann, who made me feel safe in my first year in Hamburg when the lockdown depressed me. Furthermore, I want to thank my best housemate, Yannick, he has been very warm-hearted and open-minded, and it was a great pleasure to share a roof with him during the last half a year.

Ultimately, I must acknowledge the computational service of the Deutsches Klimarechenzentrum, DKRZ. This work was supported by the degree completion grant, university of Hamburg.

Eidesstattliche Versicherung

Declaration on Oath

Ich erkläre hiermit an Eides statt, dass ich die vorliegende Masterarbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Ich habe die Arbeit nicht vorher zu einem anderen Prüfungsverfahren eingereicht. Die gedruckte Fassung ist mit der elektronischen Fassung identisch. Ich bin mit der Ausstellung der Arbeit in der Bibliothek einverstanden.

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Hamburg, den January 5, 2023

Jian, Dong